

AI-ENHANCED NETWORK TRAFFIC AND PROCESS ANALYTICS FOR THREAT DETECTION AND CYBER RESILIENCE METRIC OPTIMISATION (OT/ICS FOCUS)

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Abstract

Operational Technology (OT) and Industrial Control System (ICS) anomaly detection research has advanced substantially in detection accuracy, yet evaluation remains accuracy-only, explainability is rarely assessed quantitatively, and detections are rarely linked to threat intelligence. This dissertation addresses these gaps via an AI-driven, explainable detection framework designed and evaluated across three OT/ICS testbeds (SWaT, WaDi, BATADAL) using Design Science Research. Four models (Random Forest, CNN, Isolation Forest, LSTM Autoencoder) were trained on protocol-aware features from SWaT's CIP/EtherNet-IP traffic fused with physical data, and physical telemetry for WaDi and BATADAL. A novel Resilience-Accuracy Score (RAS) combines F1, Mean Time to Detect (MTTD), and Mean Time to Recover (MTTR) via AHP-Entropy weighting. Explainability was evaluated via SHAP faithfulness, robustness, and complexity across all twelve combinations, and detections were mapped to MITRE ATT&CK for ICS and enriched as STIX.

Results show no single model dominates detection performance across datasets: LSTM-AE is the most consistent performer, while CNN exhibits a dataset-dependent reliability gap, strong on BATADAL but saturated on SWaT and failing outright on WaDi. RAS diverges materially from F1 on BATADAL, where LSTM-AE ranks second by F1 but last by RAS due to a markedly slower detection latency, demonstrating that the most accurate model is not always the most operationally resilient. SHAP evaluation reveals Isolation Forest's explanations to be structurally unfaithful across all three datasets, while LSTM-AE produces the most stable attributions of any model. CTI enrichment

measurably increases actionability over standalone alerting. This research's principal contributions are the Resilience-Accuracy Score itself, a fully quantitative SHAP evaluation framework for ICS anomaly detection, and a working detection-to-MITRE-ATT&CK-to-STIX threat intelligence pipeline.

Keywords: OT/ICS security, anomaly detection, explainable AI, SHAP, cyber resilience, MTTD, MTTR, Resilience-Accuracy Score, cyber threat intelligence, MITRE ATT&CK.

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Glossary and List of Abbreviations

This glossary covers the abbreviations, technical terms, and dataset names used throughout this dissertation. The List of Abbreviations gives quick expansions; the Glossary of Terms gives fuller definitions specific to how each term is used in this study, with cross-references to the relevant section. Terms are listed alphabetically in both sections. Reference lists of this kind are set at 1.15 line spacing rather than the double spacing used in the main chapters; adjust if your formatting requirements specify otherwise.

List of Abbreviations

Abbreviation	Full Term
AHP	Analytic Hierarchy Process
ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge
BATADAL	BATtle of the Attack Detection ALgorithms
CI	Consistency Index
CIP	Common Industrial Protocol
CNN	Convolutional Neural Network
CR	Consistency Ratio
CRQF	Cyber Resilience Quantification Framework
CTI	Cyber Threat Intelligence
DL	Deep Learning

DNP3	Distributed Network Protocol 3
DSR	Design Science Research
EPANET	Environmental Protection Agency Network (hydraulic simulation software)
HMI	Human-Machine Interface
ICS	Industrial Control System
IEC	International Electrotechnical Commission
IF	Isolation Forest
IIoT	Industrial Internet of Things
ISA	International Society of Automation
IT	Information Technology
LIME	Local Interpretable Model-agnostic Explanations
LSTM-AE	Long Short-Term Memory Autoencoder
MI	Mutual Information
ML	Machine Learning
MTTD	Mean Time To Detect
MTTR	Mean Time To Recover
NIDS	Network Intrusion Detection System
NIST	National Institute of Standards and Technology

OASIS	Organization for the Advancement of Structured Information Standards
OPC UA	Open Platform Communications Unified Architecture
OT	Operational Technology
PLC	Programmable Logic Controller
RAS	Resilience-Accuracy Score
RF	Random Forest
RI	Random Index (Saaty's)
RQ	Research Question
SCADA	Supervisory Control and Data Acquisition
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
SMOTE	Synthetic Minority Oversampling Technique
STIX	Structured Threat Information eXpression
SUTD	Singapore University of Technology and Design
SWaT	Secure Water Treatment
TAXII	Trusted Automated eXchange of Indicator Information
WaDi	Water Distribution
XAI	Explainable Artificial Intelligence

XGBoost	eXtreme Gradient Boosting

Glossary of Terms

Actuator. A physical device (e.g., a pump or valve) in an OT/ICS environment that carries out a commanded action, as opposed to a sensor, which measures a process variable. One of three tag-type categories used to map SHAP-flagged detections to candidate MITRE ATT&CK for ICS techniques (Section 3.9.1).

AHP-Entropy Combined Weighting. The method used to derive the weights underlying the Resilience-Accuracy Score. Subjective priority weighting, from a pairwise Analytic Hierarchy Process comparison, is combined with objective, data-derived weighting from Shannon entropy, so the final weighting reflects both expert judgement and how much each criterion actually varies across the models being compared (Section 3.7.3).

Background/Reference Set. In SHAP, the set of instances an explainer compares a given prediction against to compute attributions. Drawn exclusively from the training partition in this study, never test, to avoid leakage into the explanation baseline itself (Section 3.8.3).

BATADAL. A dataset generated from a simulated water distribution network (the fictional C-Town), released through a detection-algorithm competition. Used as a physical-layer-only generalisability benchmark in this study, since it never had a network layer (Section 3.4).

Chronological Split. A train/test division that preserves time order rather than shuffling data randomly, used throughout this study to prevent future information from leaking into model training.

Complexity (SHAP). One of three quantitative XAI metrics used in this study. The Shannon entropy of an explanation's normalised attribution values; lower complexity indicates attribution concentrated on a few features, a more parsimonious explanation (Section 3.8.2).

Consistency Ratio (CR). A diagnostic value confirming whether a pairwise AHP comparison matrix reflects internally consistent judgements. Below Saaty's threshold of 0.1, a matrix is considered acceptable; this study's matrix achieved CR = 0.016 (Section 3.7.3).

Cross-Validation (Chronological Expanding-Window). The model-selection strategy used in this study: 3-fold, with each fold's boundaries anchored to attack-

row position where possible, and folds producing degenerate (single-class) validation sets excluded from selection (Section 3.6.2).

Design Science Research (DSR). The overarching research paradigm adopted for this study, appropriate for constructing and evaluating a novel artefact (the detection pipeline and its RAS metric) through iterative cycles of relevance, design, and rigour (Section 3.2).

Determinism (TensorFlow). The property that repeated runs of the same code produce identical results. An ordering bug (a determinism flag applied after TensorFlow's own import) meant this property did not hold for some early training runs; found, fixed, and disclosed (Sections 3.6.6 and 6.3).

Dual Ground-Truth Verification. The comparison, in SWaT, of the physical historian's own attack label against an independently derived network-layer attack label, used to assess data quality (Section 3.5.5).

EPANET. Hydraulic and water-quality simulation software used to generate the BATADAL dataset. BATADAL accordingly has no network layer of any kind, since it was never a captured real system.

Faithfulness (SHAP). One of three quantitative XAI metrics used in this study. The Pearson correlation, across randomly masked feature subsets, between an explanation's attributed importance and the actual resulting change in model output; a faithful explanation shows a real, proportional output change when the features it identifies as important are removed (Section 3.8.2).

Feature Selection (three-stage). The dimensionality-reduction process applied once per dataset before model training: correlation filtering, mutual information scoring against the attack label, then SHAP-guided importance ranking using a proxy Random Forest (Section 3.5.4).

Flag Fraction. The proportion of test-set rows a model is set to flag as anomalous, used in place of an absolute probability or reconstruction-error threshold, since absolute thresholds proved unstable across partitions (Section 3.6.6).

GradientExplainer. The SHAP explainer used for the CNN and LSTM Autoencoder models in this study, suited to differentiable Keras models (Section 3.8.1).

Ground Truth. The reference labels (Attack/Normal) against which model predictions are evaluated. In SWaT, verified via dual ground-truth comparison against an independently derived network-layer label (Section 3.5.5).

Historian. Industry term for the data-logging system recording physical process readings from an OT/ICS environment over time. A gap in SWaT's historian

recording, at the start and end of the network capture, produced the 110,552-row fusion boundary gap discussed in Section 3.5.5.

Isolation Forest (IF). An unsupervised, tree-based anomaly detection algorithm. Isolates anomalies via how few random partitioning splits are needed to separate an instance, rather than learning a decision boundary from labelled data. Found in this study to be structurally incompatible with SHAP's additive attribution framework (Section 3.8.4).

LSTM Autoencoder (LSTM-AE). An unsupervised deep learning architecture trained to reconstruct normal input sequences; anomalies are flagged via elevated reconstruction error. Selected for its ability to capture OT data's time-dependent structure.

Mean Time to Detect (MTTD). A resilience metric used in this study: the elapsed time from an attack event's first row to the first row a model correctly flags. Undetected events are excluded from the average and contribute instead to a separately-reported miss rate (Section 3.7.2).

Mean Time to Recover (MTTR). A resilience metric used in this study, deliberately redefined from its conventional incident-response meaning, since the pipeline performs no automated response: the elapsed time from an attack event's end to the last trailing false-positive prediction, used as a proxy for alarm-fatigue burden (Section 3.7.2).

MinMax Normalisation. A scaling method mapping values to the [0,1] range based on each dataset's own minimum and maximum. Applied to CNN and LSTM-AE inputs; Random Forest and Isolation Forest instead use unscaled features (Section 3.5.4).

Miss Rate. The proportion of attack events a model never detects at all. Applied as a multiplicative penalty on the final Resilience-Accuracy Score (Section 3.7.3).

Modbus. A widely used, legacy industrial communication protocol. Field names in SWaT's raw network capture retain “Modbus_*” naming for historical reasons, despite the underlying traffic being confirmed as CIP/EtherNet-IP, not Modbus (Section 3.5.2).

Reconstruction Error. The difference between an autoencoder's reconstructed output and its original input, used by the LSTM Autoencoder as its anomaly score, and, via a wrapper model, as the differentiable scalar SHAP explains for that architecture (Section 3.8.1).

Resilience-Accuracy Score (RAS). This study's principal novel contribution: a composite metric combining F1, MTTD, and MTTR via AHP-Entropy combined

weighting, computed independently per dataset, demonstrated to diverge materially from F1 rankings on BATADAL (Section 3.7.3).

Robustness (SHAP). One of three quantitative XAI metrics used in this study. The largest explanation change observed across several small random input perturbations (the max-sensitivity principle); lower values indicate a more stable explanation (Section 3.8.2).

Saaty Scale. The 1-9 numerical scale used in Analytic Hierarchy Process pairwise comparisons to express relative priority between two criteria.

Sensor. A physical device in an OT/ICS environment that measures a process variable (e.g., level, flow, pressure), as opposed to an actuator, which carries out a commanded action. One of three tag-type categories used in this study's CTI mapping (Section 3.9.1).

SHAP Value. The attributed contribution of an individual feature to a specific model prediction, derived from cooperative game theory (Shapley values). The basis for this study's explainability evaluation and CTI mapping.

Stratified Sampling. A sampling method guaranteeing a minimum share of the minority (attack) class in a sample, used throughout this study's SHAP evaluation to avoid drawing samples with negligible attack representation on low-attack-rate datasets (Section 3.8.3).

STIX Indicator. A structured object, following the STIX 2.1 standard, representing a single piece of threat information. Each indicator in this study carries a mapped MITRE ATT&CK technique plus custom fields recording the detecting model, its RAS score, and its top SHAP-attributed feature (Section 3.9.2).

SWaT. A six-stage water treatment testbed operated by iTrust, SUTD. The only dataset in this study providing genuine, temporally-aligned network traffic alongside physical process data from the same continuous capture, and accordingly the primary case study (Section 3.4).

TreeExplainer. The SHAP explainer used for the Random Forest and Isolation Forest models in this study, computing exact Shapley values for tree-ensemble models (Section 3.8.1).

WaDi. A water distribution network testbed operated by iTrust, SUTD, downstream of SWaT's reverse-osmosis output. Contributes physical-layer evidence only in this study, since its available network capture is temporally disjoint from its attack-labelled physical data (Section 3.4).

Windowing. The process of dividing a continuous time series into fixed-length, overlapping sequences for use by the CNN and LSTM Autoencoder models. Window length and stride are set per dataset's native sampling rate (Section 3.6.4).

Chapter 1: Introduction

1.1 Background

Industrial Control Systems (ICS) and Operational Technology (OT) environments underpin critical national infrastructure across energy, water treatment, manufacturing, and transportation (Seo et al., 2026). Historically air-gapped and prioritising availability and safety over confidentiality (Duo et al., 2022), these environments rely on proprietary protocols (Modbus, DNP3, OPC UA, CIP/EtherNet-IP) lacking authentication or encryption by default (Kelli et al., 2022; Gebhard and Perouli, 2024). IT/OT convergence, driven by IIoT adoption, cloud-based supervisory control, and remote access, has since expanded the attack surface, while OT's constraints (low latency tolerance, high availability, safety-critical operation) preclude directly applying conventional IT security tools (Duo et al., 2022; Alem et al., 2023; Kelli et al., 2022).

High-profile incidents, from Stuxnet to the Colonial Pipeline ransomware attack, confirm the severity of this threat landscape (Duo et al., 2022; Seo et al., 2026; CISA, 2022), exploiting industrial protocols and cyber-physical constraints (Halbouni et al., 2022; examined further in Section 2.2). OT data is also inherently multivariate and time-dependent, demanding detection approaches distinct from IT network traffic (Section 2.2.3).

1.2 Problem statement

Applying machine learning (ML) and deep learning (DL) to ICS/OT anomaly detection presents six interrelated problems motivating this research. First, a tension exists between accuracy and interpretability: supervised models (Random Forest, CNN) achieve high accuracy but remain opaque, while unsupervised models (Isolation Forest, LSTM Autoencoder) detect novel attacks with even less interpretive context (Macas, Wu and Fuertes, 2022; Kravchik and Shabtai, 2022), undermining operator trust (Halbouni et al., 2022). Second, XAI applied to ICS remains nascent, targeting IT rather than OT scenarios (Nwakanma et al., 2023; Kalakoti et al., 2025). Third, evaluation relies on accuracy-oriented metrics alone, without MTTD/MTTR, and no existing work combines both into a single ranking (Alem et al., 2023; Kelli et al., 2022). Fourth, many works contribute a survey or dataset but not a complete end-to-end pipeline (Duo et al., 2022; Halbouni et al., 2022). Fifth, detected anomalies are rarely mapped to threat intelligence frameworks such as MITRE ATT&CK for ICS, leaving detection reactive (Hossain et al., 2024; Pirca and Lallie, 2023). Sixth, validation frequently relies on generic IT datasets, limiting real-world transferability (Kelli et al., 2022; Alem et al., 2023).

1.3 Research gaps

A systematic analysis of ICS/OT anomaly detection literature reveals six distinct, interrelated gaps justifying this research.

Gap 1 (Explainability): most ICS studies deploy ML/DL without explainability, and where XAI is applied it is rarely evaluated using quantitative metrics such as

faithfulness, robustness, and complexity (Nwakanma et al., 2023; Macas, Wu and Fuertes, 2022; Moustafa et al., 2023). This project addresses Gap 1 via SHAP-based explainability across all trained models, evaluated against quantitative XAI metrics (Kalakoti et al., 2025).

Gap 2 (Resilience metrics): existing work evaluates detection performance solely through accuracy metrics, omitting MTTD and MTTR, which directly determine physical disruption and safety consequences (Alem et al., 2023; Lubis et al., 2025). This project addresses Gap 2 by deriving MTTD/MTTR from model outputs and incorporating them into the evaluation framework.

Gap 3 (OT-specific context): many works do not target OT-specific protocols or account for OT constraints such as low latency and safety-critical operation (Kelli et al., 2022; Gebhard and Perouli, 2024; Alem et al., 2023). This project addresses Gap 3 through OT-relevant datasets (SWaT, WaDi, BATADAL) and protocol-aware feature engineering (Goh et al., 2017; Ahmed, Palleti and Mathur, 2017).

Gap 4 (End-to-end pipeline): many works contribute a review or dataset but not a complete, implemented pipeline (Duo et al., 2022; Halbouni et al., 2022). This project addresses Gap 4 by implementing a full pipeline from data ingestion through XAI output.

Gap 5 (CTI integration): detected events are rarely mapped to MITRE ATT&CK for ICS or STIX/TAXII, leaving detection reactive rather than proactive (Hossain et al., 2024; Pirca and Lallie, 2023). This project addresses Gap 5 by enriching detections with CTI context.

Gap 6 (Dual evaluation): every reviewed work evaluates either accuracy or resilience, never both, and none ranks models by a combined score (Alem et al., 2023; Kelli et al., 2022). This project addresses Gap 6 by introducing the Resilience-Accuracy Score (RAS), demonstrating that the most accurate model is not always the most operationally resilient.

1.4 Aim and objectives

1.4.1 Aim

This research aims to design, implement, and evaluate an AI-driven, explainable anomaly detection framework for ICS integrating protocol-aware network-and-process feature engineering, supervised and unsupervised ML/DL models, SHAP explainability, CTI enrichment, and a novel Resilience-Accuracy Score. The framework is developed and validated primarily on SWaT, the only dataset providing genuine, temporally-aligned network traffic alongside physical process data; WaDi and BATADAL test the generalisability of the pipeline and resilience metric across distinct OT environments without contributing network-traffic evidence.

1.4.2 Objectives

1. Synthesise the ICS/OT anomaly detection literature to identify recurring limitations across explainability, resilience measurement, protocol-awareness, pipeline completeness, threat intelligence integration, and dual evaluation (Duo et al., 2022; Halbouni et al., 2022).
2. Acquire and engineer OT-contextualised features from SWaT, WaDi, and BATADAL, incorporating CIP/EtherNet-IP network-layer attributes from

SWaT and physical process dynamics across all three (Goh et al., 2017; Ahmed, Palleti and Mathur, 2017; Gebhard and Perouli, 2024).

3. Design, implement, and train Random Forest, CNN, Isolation Forest, and LSTM Autoencoder via grid-search hyperparameter optimisation (Kravchik and Shabtai, 2022; Macas, Wu and Fuertes, 2022).
4. Apply SHAP across all trained models to generate feature-level attributions, evaluating explanation quality using quantitative XAI metrics (Kalakoti et al., 2025; Nwakanma et al., 2023).
5. Derive MTTD and MTTR and compute the Resilience-Accuracy Score using AHP-Entropy weighting, combining expert-judgement weighting with data-driven objectivity (Zhou and Chen, 2025).
6. Map detected anomalies to MITRE ATT&CK for ICS techniques and enrich detections with CTI indicators (Hossain et al., 2024; Pirca and Lallie, 2023).
7. Integrate all components into a complete pipeline and evaluate it against all six identified gaps.

1.5 Research questions

RQ1: How effectively do supervised (Random Forest, CNN) and unsupervised (Isolation Forest, LSTM Autoencoder) models identify attacks across SWaT, WaDi, and BATADAL, measured by precision, recall, and F1-score?

RQ2: What are the comparative strengths and limitations of supervised versus unsupervised approaches across datasets?

RQ3: To what extent do SHAP explanations improve interpretability and trustworthiness, per quantitative XAI metrics and alignment with ground-truth attack characteristics?

RQ4: How do models compare under the proposed Resilience-Accuracy Score, and does the most accurate model consistently rank as most operationally resilient?

RQ5: To what extent does CTI integration, including MITRE ATT&CK mapping and STIX/TAXII enrichment, enhance actionability compared to standalone alerting?

1.6 Scope, Delimitations, and Assumptions

1.6.1 Scope

This research spans data ingestion to operator-facing outputs (Duo et al., 2022), using SWaT (Goh et al., 2017), WaDi (Ahmed, Palleti and Mathur, 2017), and BATADAL (Taormina et al., 2018); no personal or commercially sensitive data are processed. Only SWaT provides network-layer traffic temporally aligned with its labelled physical attack data; WaDi's 2023 network capture is disjoint from its 2017 attack-labelled data and is excluded, and BATADAL, an EPANET simulation, never had a network layer. SWaT is accordingly the primary network-traffic case study; WaDi and BATADAL serve as physical-layer-only generalisability benchmarks. Four models spanning the supervised/unsupervised and traditional/deep-learning spectrum are evaluated (Kravchik and Shabtai, 2022; Macas, Wu and Fuertes, 2022), with protocol-aware feature engineering targeting CIP/EtherNet-IP traffic in the SWaT captures, and WaDi/BATADAL feature engineering confined to physical process and SCADA tag-level attributes. SHAP-based explainability uses quantitative XAI metrics (Kalakoti et

al., 2025); MTTD/MTTR combine with accuracy through RAS (Alem et al., 2023; Lubis et al., 2025); detections are mapped to MITRE ATT&CK for ICS and enriched with STIX/TAXII indicators (Hossain et al., 2024).

1.6.2 Delimitations

Experiments are conducted offline on pre-collected testbed data, excluding live deployment. Model selection is bounded to the four architectures above, with Transformer- and GNN-based approaches discussed in Chapter 2 but excluded from implementation. Protocol coverage is representative rather than exhaustive. CTI integration is demonstrated at the design level only, without real-time feed integration or automated response. Adversarial robustness testing and formal operator user studies are not conducted; explanation quality is assessed solely through quantitative XAI metrics. Network-traffic analytics is validated only on SWaT; findings do not extend to WaDi or BATADAL, evaluated solely on physical/process-layer features.

1.6.3 Assumptions

SWaT, WaDi, and BATADAL are assumed to sufficiently approximate real-world OT behaviour for meaningful evaluation (Goh et al., 2017; Ahmed, Palleti and Mathur, 2017; Taormina et al., 2018). Ground-truth attack labels are assumed accurate and complete. Protocol-aware features are assumed to capture sufficiently general OT communication semantics (Kelli et al., 2022; Gebhard and Perouli, 2024). SHAP is assumed to provide reliable attributions for the selected architectures on ICS time-series data (Kalakoti et al., 2025). MTTD/MTTR and MITRE ATT&CK mappings are assumed derivable from the datasets, acknowledging testbed scenarios may not fully replicate real-world advanced persistent threats.

1.7 Method overview

This research adopts Design Science Research (vom Brocke, Hevner and Maedche, 2020), appropriate for constructing and evaluating an artefact, the AI-enhanced OT IDS pipeline and its RAS metric, through iterative cycles of problem identification, design, implementation, and evaluation. The experimental design proceeds through four stages, detailed fully in Chapter 3: protocol-aware feature engineering; multi-model training (Random Forest, CNN, Isolation Forest, LSTM Autoencoder) with cross-dataset evaluation; SHAP-based explainability; and dual accuracy-resilience evaluation via the Resilience-Accuracy Score with CTI-enriched MITRE ATT&CK mapping.

Chapter 2: Literature Review

2.1 Introduction

This chapter examines existing scholarship on anomaly detection in ICS/OT environments, progressing from domain context and threat landscape, through datasets and ML/DL detection methods, to explainable AI, evaluation frameworks, cyber resilience, and CTI integration, before synthesising six research gaps. All sources are from 2022 onwards, consistent with this research's temporal scope, excepting foundational dataset-origin literature.

2.2 ICS/OT Domain Context and Threat Landscape

ICS and OT environments form the operational backbone of the critical infrastructure sectors introduced in Section 1.1, combining physical and digital elements requiring accurate process analysis and reliable SCADA communication (Gaggero et al., 2024). Historically air-gapped and prioritising availability and safety over confidentiality (Duo et al., 2022), these systems rely on protocols such as Modbus, DNP3, OPC UA, and CIP/EtherNet-IP lacking authentication or encryption (Kelli et al., 2022; Gebhard and Perouli, 2024). NIST notes OT's growing resemblance to IT as it adopts corporate connectivity and remote access, reducing isolation (NIST, 2023). The Purdue Reference Model's five levels, extended by IIoT heterogeneity in vendors, protocols, and cloud services, have widened the industrial attack surface considerably (Termanini et al., 2026; Duo et al., 2022; Hossain et al., 2024).

A critical IT/OT distinction lies in operational priorities: OT cybersecurity prioritises integrity and availability, followed by confidentiality, with safety an overarching concern that can conflict with security in system design (NIST, 2023), creating constraints (low latency tolerance, high availability, safety-critical operation) that preclude directly applying conventional IT tools (Duo et al., 2022). IEC 62443 addresses these requirements, defining roles for asset owners, integrators, and manufacturers, and security levels appropriate to the threat environment (ISA/IEC, 2024).

High-profile incidents, Stuxnet, Industroyer/CrashOverride, Triton/TRISIS, and the Colonial Pipeline ransomware attack, demonstrate that ICS/OT compromise can cause real-world physical damage (Termanini et al., 2026; Seo et al., 2026; Duo et al., 2022; CISA, 2022). Firoozjaei et al. (cited in Wang et al., 2026) analysed six such events, proposing an impact-weighting method. NIST identifies threat sources spanning hostile states, insiders, and human error, with consequences including blocked information flow and interference with safety systems (NIST, 2023).

ICS vulnerability is rooted in protocols lacking authentication, encryption, or integrity checks, enabling eavesdropping, flooding, and packet modification (Zhou et al., 2026), with common attacks including reconnaissance, DoS/DDoS, False Data Injection, and Man-in-the-Middle (Zhou et al., 2026; Gaggero et al., 2024). OT traffic is inherently multivariate and time-series, with anomalies often manifesting as subtle deviations in physical process variables (Goh et al., 2017; Taormina et al., 2018), demanding protocol-aware feature engineering rather than generic network-flow features (Kelli et al., 2022; Gebhard and Perouli, 2024; Alem et al., 2023).

2.3 ICS Security Standards and Maturity Frameworks

The IEC 62443 series provides a comprehensive industrial cybersecurity framework, grouped into General, Policies and Procedures, System, and Component categories, defining roles for Asset Owners, System Integrators, and Product Manufacturers, and Security Levels benchmarking component security (ISA/IEC, 2024). Maturity levels, based on the Capability Maturity Model Integration framework, link process to product security capability: IEC 62443-4-2 compliance requires Maturity Level 3 under 62443-4-1 (ISA/IEC, 2024).

More broadly, maturity models give organisations a structured means to assess security capability, serving descriptive, comparative, and prescriptive purposes; maturity in business continuity management correlates with organisational resilience, reinforced during COVID-19 (Brezavsek and Baggia, 2025), and information security risk has shifted from an IT concern to a board-level business risk (Brezavsek and Baggia, 2025). These frameworks address organisational posture broadly, not real-time ICS anomaly detection and response specifically.

2.4 Datasets for ICS Anomaly Detection

This research draws on three interconnected iTrust testbed datasets. SWaT is a scaled-down, six-stage water treatment testbed comprising seven days of normal operation and a six-day attack phase with 36 multi-point attacks targeting actuators, sensors, and the network, confirmed against the accompanying attack list (Goh et al., 2017). WaDi extends this downstream to a water distribution network connected to SWaT's reverse-osmosis output, controlled across three PLC-managed grids, capturing 14 days of

normal operation and 2 days of attack scenarios across 123 sensors and actuators (Ahmed, Palleti and Mathur, 2017). BATADAL, developed for the C-Town network (obtained via iTrust but produced independently for the BATADAL competition), provides SCADA observations from a real-world-scale water distribution system, released through a competition evaluating detection time and classification accuracy (Taormina et al., 2018). SWaT provides protocol-realistic network-layer data alongside physical telemetry from the same capture; WaDi provides physical telemetry with attack labels only, its network capture recorded separately in 2023; BATADAL provides SCADA tag-level time-series data without any network layer. Together these span treatment, distribution, and simulated system-level processes, though network-layer features are available only for SWaT.

Dataset quality and representativeness are fundamental to ML pipeline validity (Termanini et al., 2026), whose taxonomic analysis of 23 ICS-specific datasets across seven dimensions and a five-criterion scoring rubric finds that feature engineering quality substantially affects detection performance and that hybrid feature-selection approaches help address the high dimensionality and class imbalance common to ICS data.

The ICS-NAD dataset provides large-scale, real-world traffic from ten ICS brands and hundreds of PLCs across ten industrial processes, covering twenty attack types across four categories with sixty extracted features (Zhou et al., 2026). ICS-ADD similarly captures a complete attack lifecycle via a testbed combining SCADA, PLC, firewall, NIDS, and SIEM components, pairing raw traffic with monitoring-system logs (Gaggero et al., 2024). Mubarak et al. (2022) describe a portable ICS test kit simulating

wastewater treatment via PLC/HMI control, addressing the financial barriers limiting many universities' access to realistic ICS testbeds. Despite these advances, much research continues to rely on generic IT datasets such as NSL-KDD and UNSW-NB15 (Gaspar et al., 2024; Hossain et al., 2024), which do not capture the protocol-specific characteristics, temporal dependencies, or physical process constraints defining ICS/OT environments (Kelli et al., 2022; Alem et al., 2023).

Even among ICS-specific datasets, fidelity varies considerably: BATADAL's purely simulated EPANET origin and SWaT's physically instrumented testbed represent markedly different levels of realism, a distinction rarely made explicit when datasets are compared in aggregate (Termanini et al., 2026). This study treats that distinction as a first-class methodological consideration rather than a footnote, scoping network-traffic claims to SWaT alone and treating WaDi and BATADAL strictly as physical-layer generalisability benchmarks (Section 3.4).

2.5 Machine Learning and Deep Learning Approaches for Anomaly Detection

Traditional supervised algorithms, Random Forest, SVM, KNN, Decision Trees, Naive Bayes, remain widely used, with RF and XGBoost favoured for efficiently handling large, heterogeneous datasets at lower cost than neural networks (Hossain et al., 2024; Gaspar et al., 2024; Genuario et al., 2024), though these depend on manual feature engineering and generalise poorly to novel attacks (Morshedi and Matinkhah, 2025; Hossain et al., 2024; Genuario et al., 2024). Wang et al. (2026) compare XGBoost, LightGBM, Extra Tree, Decision Tree, and Random Forest on ICS-Flow and HAI,

finding broadly equivalent results even with reduced training data; Hossain et al. (2024) combine a deep autoencoder for feature selection with Random Forest, achieving near-perfect cross-validated performance, though without considering operational resilience (Alem et al., 2023). This near-perfect result should be read cautiously: cross-validation within a single dataset does not test whether the same architecture generalises across OT environments with different sampling rates or protocols, a limitation directly relevant to the cross-dataset evaluation undertaken in this study (Section 3.7.4), and one this study's own results speak to directly, given the same Random Forest architecture performs unevenly across SWaT, WaDi, and BATADAL (Chapter 4).

Deep learning has driven substantial gains in representational capacity (Hossain et al., 2024; Morshedi and Matinkhah, 2025). CNNs have been adapted by converting traffic into image-like representations (Wang and Ghaleb, 2023), with ICS-specific multi-CNN fusion models outperforming traditional ML (Xi et al., 2024). LSTM and RNN architectures are widely used for sequential modelling (Hossain et al., 2024; Xi et al., 2024; Dash et al., 2025), including PSO/JAYA/SSA-optimised LSTM models (Dash et al., 2025) and an LSTM-based VAE-GAN architecture avoiding costly latent-space mapping at detection time (Fan et al., 2023). Autoencoder variants remain central to unsupervised detection: a two-step classifier combining One-Class SVM with an autoencoder trains solely on benign traffic (Catillo et al., 2023), though reconstruction-error-based detection can be undermined when anomalies resemble normal patterns (Catillo et al., 2023); ARCADE, an adversarially regularised convolutional autoencoder, addresses limited-label environments (Lunardi et al., 2023). Transformer-based IDS-MTran achieves state-of-the-art results on NSL-KDD, CIC-DDoS2019, and

UNSW-NB15 (Xi et al., 2024), while hybrid CNN-LSTM-Transformer and CNN-BiLSTM architectures show further gains (Wang et al., 2025; Wang et al., 2024), and AutoSHARC combines ensemble feature selection with a lightweight CNN-Attention-GRU model in a closed-loop, explainability-driven pipeline (Farooqui et al., 2025).

Graph Neural Networks capture topological inter-dependencies that sample-independent methods miss; GNN-IDS represents networks as attack graphs augmented with real-time measurements, achieving strong performance and adversarial robustness (Sun, Teixeira and Toor, 2024). Separately, labelled-data scarcity has driven unsupervised, drift-aware approaches: an online system explicitly manages recurring concept drift by retaining prior models for reuse rather than discarding them (De Paola et al., 2024). Federated Learning has also emerged as a privacy-preserving direction for collaborative IDS (Belenguer, Pascual and Navaridas, 2025), though its ICS application remains limited by inconsistent evaluation standards.

2.6 Explainable Artificial Intelligence (XAI) in Intrusion Detection

The accuracy-interpretability tension from Section 1.2 is well-documented: supervised models are accurate but opaque, unsupervised models detect novel attacks with even less interpretive context (Macas, Wu and Fuertes, 2022; Kravchik and Shabtai, 2022), undermining operator trust (Halbouni et al., 2022). SHAP and LIME address this, SHAP offering global feature importance, LIME local instance-specific explanation, together covering both the model's overall behaviour and any single prediction (Gaspar et al., 2024); however, XAI methods carry computational cost and remain vulnerable to adversarial manipulation of explanations without altering predictions (Gaspar et al.,

2024). More fundamentally, few of these applications examine whether an explanation method's own assumptions are appropriate for the underlying detector's decision mechanism. SHAP's additive feature-attribution model, for instance, assumes a decomposable relationship between features and output; this study tests that assumption directly against all four model architectures and finds it structurally violated for Isolation Forest specifically (Section 3.8.4), a finding the reviewed literature does not anticipate.

Application to ICS specifically remains nascent, largely confined to IT contexts and, where applied, limited to feature-importance rankings without systematic evaluation against faithfulness, robustness, and complexity (Nwakanma et al., 2023; Moustafa et al., 2023; Kalakoti et al., 2025). Fung et al. (2024) evaluate five detection models and eight attribution methods across real and synthetic ICS attacks: ranking features by raw reconstruction error performs poorly across a broad attack set; ML-based attribution outperforms this baseline only at anomaly onset; ensemble attribution outperforms any single method; low-magnitude, categorical-actuator manipulations remain hardest to attribute, directly challenging the sufficiency of reconstruction-error-based guidance.

Feedback-driven approaches point toward integration rather than post-hoc application: AutoSHARC embeds SHAP into a closed retraining loop reducing overhead and improving transparency (Farooqui et al., 2025), while IoT-focused work shows feature-level interpretability supports auditing and operational trust beyond accuracy reporting alone (Morshedi and Matinkhah, 2025).

2.7 Evaluation Frameworks and Operational Resilience Metrics

A persistent limitation is reliance on accuracy-oriented metrics, precision, recall, F1-score, which do not capture operational impact in safety-critical ICS deployment (Wang et al., 2026; Hossain et al., 2024; Alem et al., 2023). None of the reviewed evaluations compute resilience indicators such as MTTD or MTTR, despite these directly determining potential physical damage (Lubis et al., 2025). Morshedi and Matinkhah (2025) go further with a Pareto analysis of accuracy against latency, but remain at the model level rather than system-level operational resilience. No existing work ranks models on a combined accuracy-resilience score (Alem et al., 2023), a meaningful gap, since a marginally less accurate but faster model may be operationally preferable.

Cyber resilience more broadly denotes a system's capacity to prepare for, absorb, recover from, and adapt to attacks (AlHidaifi, Asghar and Ansari, 2024). The Cyber Resilience Quantification Framework models this via five factors, managing complexity, network topology, added resources, pre-configuration, and buffering support, using simulation rather than formal testing (AlHidaifi, Asghar and Ansari, 2024); however, it targets IT infrastructure broadly and has not been applied to ICS anomaly detection evaluation. The CRQF's five factors are also validated only through simulation rather than against real detector outputs, leaving open whether a similarly structured composite score behaves the same way when its inputs come from genuinely trained ML models rather than simulated system states. This is precisely the gap the AHP-Entropy RAS construction in this study addresses empirically, combining subjective and data-derived weighting over real F1, MTTD, and MTTR values rather than simulated ones (Section 3.7.3).

2.8 Cyber Threat Intelligence Integration

A further gap is the disconnect between anomaly detection and CTI: detected events are rarely mapped to frameworks such as MITRE ATT&CK for ICS or STIX/TAXII, leaving detection reactive rather than proactive (Hossain et al., 2024; Pirca and Lallie, 2023). This limits operational value, since teams cannot readily contextualise alerts or share intelligence externally. IEC 62443 identifies communication and cooperation as a core requirement (ISA/IEC, 2024) and NIST guidance stresses detecting events before escalation (NIST, 2023), yet neither addresses translating outputs into shareable intelligence. Analysis of six major real-world events shows attacks follow identifiable patterns mappable to threat frameworks (Firoozjaei et al., cited in Wang et al., 2026), a mapping largely absent from current systems.

2.9 Identification of Research Gaps

The preceding review confirms six interrelated gaps. Gap 1 (explainability) is confirmed by the near-total absence of XAI in ICS-specific studies (Nwakanma et al., 2023; Macas, Wu and Fuertes, 2022); where XAI does appear, it is confined to IT contexts (Gaspar et al., 2024; Farooqui et al., 2025) or performs poorly when transferred to ICS without adaptation (Fung et al., 2024).

Gap 2 (resilience metrics) is exemplified by Wang et al. (2026) and Hossain et al. (2024), both reporting classification metrics with no operational context; AlHidaifi, Asghar and Ansari (2024) offer a resilience framework, but for IT infrastructure rather than ICS detection.

Gap 3 (OT-specific context) follows from continued reliance on generic IT datasets (Gaspar et al., 2024); ICS-specific datasets are emerging (Zhou et al., 2026; Gaggero et al., 2024; Mubarak et al., 2022), but protocol-aware feature engineering is rarely built on top of them.

Gap 4 (end-to-end pipelines) is evident in how contributions remain siloed: feature selection (Farooqui et al., 2025), detection (Wang et al., 2026), and attribution (Fung et al., 2024) are each addressed in isolation.

Gap 5 (CTI integration) holds across the systems reviewed; none map outputs to MITRE ATT&CK for ICS or enrich them via STIX/TAXII (Hossain et al., 2024; Pirca and Lallie, 2023).

Gap 6 (dual evaluation) is the most consistent finding: every source evaluates accuracy or resilience, never both together (Alem et al., 2023). Morshedi and Matinkhah (2025) come closest with a Pareto analysis, but stop short of a formal combined framework.

Table 2.1: *Gap coverage across core comparable works.*

Work	Focus	Gap(s) Addressed
Mishra et al. (2025)	Ensemble/DL anomaly detection benchmarking in ICS	1, 2, 5, 6
Shewale (2023)	Survey of ML/DL for ICS anomaly detection	4, 6

Dehlaghi-Ghadim et al. (2023)	Purpose-built ICS anomaly detection dataset	2, 4, 6
Nguyen et al. (2024)	ML/DL benchmarking on HAI and ICS-Flow	1, 2, 5, 6
Oyedotun et al. (2025)	Multimodal DL and XAI for SCADA/DCS	6
Gaggero et al. (2024)	ICS-ADD dataset for cyber-physical monitoring	4, 6
Benka et al. (2025)	CNN-based detection on simulated PLC attacks	1, 2, 5, 6
Mubarak et al. (2022)	ICS testbed and ML-based attack detection	2, 6
Sun et al. (2023)	CTI mining survey for proactive defence	4, 6
Presekal et al. (2023)	Attack graph and hybrid DL for power systems	5, 6
AlHidaifi, Asghar and Ansari (2024)	IT cyber resilience quantification framework	6

Skrodelis and Romanovs (2026)	Review of hybrid IDS for SCADA/ICS	4, 6
Morshedi and Matinkhah (2025)	Review of DL for IoT anomaly detection	4, 6

No reviewed work addresses more than four of the six gaps simultaneously, and Gap 6 (dual accuracy-resilience evaluation) is absent from every work in this table, the most consistent finding across the wider 27-work review underlying this chapter.

These gaps are not independent: weak explainability (Gap 1) makes CTI mapping harder (Gap 5), accuracy-only evaluation (Gap 2) forecloses dual evaluation (Gap 6), reliance on generic datasets (Gap 3) limits generalisability, and without an integrated pipeline (Gap 4), no system closes all six at once.

2.10 Summary

This review has examined ICS/OT anomaly detection across domain context, security standards, datasets, ML/DL approaches, explainable AI, evaluation and resilience metrics, and CTI integration, tracing detection accuracy gains from traditional classifiers (Hossain et al., 2024; Wang et al., 2026) through LSTM networks (Dash et al., 2025; Fan et al., 2023), Transformers (Xi et al., 2024), GNNs (Sun, Teixeira and Toor, 2024), and hybrid models (Farooqui et al., 2025). The six gaps established in Section 2.9 remain interrelated and unaddressed in combination by any single reviewed work; Chapter 3 presents the Design Science Research methodology designed to close them.

Chapter 3: Methodology

3.1 Introduction

This chapter sets out the methodology addressing the six gaps and objectives from Chapters 1-2: Design Science Research (3.2), the research-question-to-method mapping (3.3), dataset provenance (3.4), pre-processing through CTI integration (3.5-3.9), and ethics (3.10). Full implementation detail and forensic narratives are in Appendix A.

3.2 Research Philosophy and Design Science Research

This study constructs and evaluates a novel artefact, an explainable multi-model pipeline with a bespoke Resilience-Accuracy Score, rather than testing pre-existing theory, motivating Design Science Research (vom Brocke, Hevner and Maedche, 2020) over a purely experimental design: relevance (Chapter 2's six gaps), design (feature engineering, multi-model training, SHAP, RAS/CTI evaluation), and rigour (validated ICS-native datasets, Section 3.4).

3.3 Mapping Research Questions to Methods and Metrics

Table 3.1 maps each research question to its method and metric, following Kim et al.'s (2023) cross-model comparison protocol.

Table 3.1: *Research question to method and metric mapping.*

Research Question	Method / Model	Evaluation Metric
RQ1: supervised vs unsupervised detection	Multi-model training and testing (3.6)	Precision, Recall, F1 (3.7.1)
RQ2: comparative strengths across datasets	Cross-dataset evaluation protocol (3.7.4)	F1 variance across datasets
RQ3: SHAP interpretability gain	SHAP application and XAI evaluation (3.8)	Faithfulness, robustness, complexity
RQ4: accuracy vs RAS ranking	RAS via AHP-Entropy weighting (3.7.3)	RAS score; F1-RAS rank correlation
RQ5: CTI actionability gain	Detection-to-technique mapping, STIX (3.9)	Structured actionability assessment

3.4 Dataset Provenance and Justification

Three datasets were obtained via iTrust, SUTD (Table 3.2). SWaT (Goh et al., 2017) alone has genuine, temporally-aligned network traffic (CIP/EtherNet-IP, not Modbus despite legacy "Modbus_*" naming) alongside physical data at one-second resolution, and is the primary case study. WaDi (Ahmed, Palleti and Mathur, 2017) contributes physical evidence only, its 2023 network capture excluded as temporally disjoint from

2019 attack labels. BATADAL (Taormina et al., 2018), EPANET-simulated, never had a network layer. Both serve as generalisability benchmarks; a consistent chronological, non-random split prevents leakage across all three (Section 3.7.4).

Table 3.2: *Dataset characterisation.*

Dataset	Origin	Files supplied	Structure	Ground truth
SWaT	iTrust physical testbed	Normal_v0/v1, Attack_v0, network logs	7 days normal + 6-day attack (36 attacks); historian and network data throughout	Embedded labels, cross-verified
WaDi	iTrust physical testbed	WADI_14days_new, WADI_attackdataLABLE	14 days normal + 2 days attack; 123 sensors/actuators, 3 PLC grids	Timestamped labels
BATADAL	Simulated C-Town network	dataset03, dataset04	dataset03: 1yr normal; dataset04: ~6mo, 7 attacks	ATT_FLAG in dataset04

3.5 Pre-processing and Protocol-Aware Feature Engineering

3.5.1 Scope and rationale

Feature engineering differs by dataset based on genuinely available modalities (Table 3.3). Protocol-aware feature engineering (Kelli et al., 2022) applies only to SWaT, the sole dataset with a usable, label-alignable network capture. SWaT and WaDi's network captures both show CIP over EtherNet/IP (Gebhard and Perouli, 2024), evidenced by Rockwell Automation device identifiers.

Table 3.3: *Feature engineering scope by dataset.*

Dataset	Scope
SWaT	Network + physical (same continuous December 2015 run, joinable by timestamp)
WaDi	Physical only (2023 network capture temporally disjoint from 2019 attack data)
BATADAL	Physical only (purely simulated; no packet-level data ever generated)

3.5.2 SWaT: network and physical feature fusion

SWaT's network capture, pre-parsed by iTrust, fully overlaps both physical files. Records carry a timestamp, host/protocol/application identifiers, CIP service/value fields, SCADA tag, and port, aggregated into a per-second, per-tag feature set (packet_count, request_count, response_count, read_write_ratio, avg_value,

txn_repeat_count, txn_max_gap corrected for 16-bit wraparound, net_label_second) and fused with physical readings by timestamp (full implementation in Appendix A).

3.5.3 WaDi and BATADAL: physical/process-layer feature engineering

WaDi features are drawn from 123 sensors/actuators across three PLC-managed grids; no network-layer features are engineered, a deliberate scope decision. BATADAL features are drawn directly from its SCADA tag columns (tank levels, pump/valve status, flow rates); no network layer exists at any stage.

3.5.4 Common pre-processing procedures

Across all three datasets: missing and duplicate timestamps are resolved and dropout flagged rather than imputed; labels are joined to features by timestamp; CNN and LSTM-AE inputs are min-max normalised while Random Forest and Isolation Forest use unscaled features (Kravchik and Shabtai, 2022); fixed-length windows are built for CNN/LSTM-AE (Section 3.6.4); and a chronological, non-random split is applied throughout.

A three-stage feature-selection process was applied once per dataset on the training partition only, then applied identically to test, with all four models per dataset sharing the same selected feature set. Stage 1 (correlation filtering) drops the second of any feature pair exceeding a Pearson-r threshold, deterministically, before the costlier stages run on a smaller set. Stage 2 (mutual information scoring) ranks surviving features by mutual information with the label and retains the top-scoring subset. Stage 3 (SHAP-guided importance ranking) trains a proxy Random Forest on the Stage 2 subset, computes SHAP values, and retains only features whose mean absolute SHAP

value meets a minimum fraction of the maximum. Stages 2 and 3 subsample large training partitions (SWaT: approximately 845,000 rows) for tractability, since mutual information scoring and SHAP's TreeExplainer both scale poorly with row count; Stage 1 runs on the full training partition.

3.5.5 Network-Physical Fusion: Implementation and Data Quality

SWaT's network capture (786 parsed files, 13 date folders, 21/22 December 2015 to 3 January 2016) was fused with the physical historian via a nearest-timestamp join (one-second tolerance), producing 1,055,475 rows (full implementation in Appendix A). Verification surfaced and corrected four defects: one mis-dated row (5 December instead of 25 December 2015); twelve duplicate boundary-second records, merged by summing counts; a delimiter-shift defect displacing a network attack flag; and a fusion boundary gap of 110,552 rows (10.47%) where the historian had not yet, or no longer, been recording. These rows were dropped rather than imputed, giving a train/test split of 755,938 rows (44,086 attacks, 5.83%) and 188,985 rows (9,763 attacks, 5.17%).

Dual ground-truth verification found 99.91% agreement (922 of 1,055,475 disagreements): 21.7% (200) reflected short cross-layer detection lag at attack boundaries, and the remaining 722 comprised one isolated second (resolved above) and one unbroken 721-second run (29 December 2015, 06:30-06:42) where the physical label showed Attack with no discrete actuator change. Staged investigation (Appendix A) found only FIT601 genuinely sustained (278 of 721 seconds, from 06:30:00); other flagged tags showed brief 15-29s holds, and tank-level tags were cyclical artefacts. As no tag held the full window, the block is documented as an investigated, unresolved limitation, not a confirmed attack.

Table 3.4: *Dual ground-truth disagreement progression.*

Stage	Result
Initial comparison	922 / 1,055,475 disagreements (99.91% agreement)
Classified against attack windows	92 inside window; 20 within 5s; 88 within 6–60s; 722 more than 60 min away
Structural analysis of the 722	1 isolated second; 1 run of 721 consecutive seconds
After correcting the isolated second	921 remain; 721-second block unresolved

3.5.6 WaDi and BATADAL: Implementation and Data Quality

A diagnostic script mirroring the SWaT checks (Appendix A) was applied to WaDi and BATADAL's physical files. BATADAL required no correction: dataset03 (8,761 rows) was clean; dataset04 (4,177 rows) had a column-naming defect (whitespace in the ATT_FLAG header) silently excluding all 219 attack rows on concatenation, traced against the documented 219-hour total and corrected at source.

WaDi required more reconstruction: a stray header-offset row in the attack file, fixed by skipping the first line, and a corrupted, hour-less Time field in both files (explaining an initial 82.78% date-parse failure). Timestamps were rebuilt from each row's date plus row position (1 Hz assumed), uncovering a genuine ~73.3-hour (264,001-row) acquisition gap in the normal file (30 September-1 October 2017), confirmed via the

raw row-index field. The 26,401 post-gap rows were right-aligned to precede the validated start of 3 October (best-supported, not independently verified); the result spans 25 September-7 October 2017, zero duplicates (784,571 rows). The attack file showed no comparable gap, yielding a clean 172,801-row file (9-11 October 2017), 5.77% attack rate.

Table 3.5: *WaDi and BATADAL summary.*

File	Rows	Outcome
BATADAL dataset03	8,761	No defects; verified clean
BATADAL dataset04	4,177	ATT_FLAG header whitespace defect; corrected at source
WaDi normal	784,571	Time field, date format, and ~3-day gap defects; reconstructed and validated
WaDi attack	172,801 valid	Header offset defect; reconstructed and validated

3.6 Model Development and Training

3.6.1 Model Selection Rationale

Four models span the supervised/unsupervised and traditional/deep-learning axes flagged in Chapter 2, enabling RQ1/RQ2's paradigm comparison. Random Forest suits

large, heterogeneous ICS datasets cheaply (Wang et al., 2026; Hossain et al., 2024), its feature importances sense-checking Section 3.8's SHAP output. CNN suits the windowed representation (3.5.4), outperforming traditional ML on comparable tasks (Xi et al., 2024). Isolation Forest needs no attack labels, addressing label scarcity, and is mechanistically distinct from LSTM-AE. LSTM-AE suits OT data's multivariate, time-dependent nature; its degradation on normal-like anomalies (Catillo et al., 2023) motivates RAS (3.7.3), central to Gap 6. Transformer/GNN architectures are excluded per Section 1.6.2.

3.6.2 Cross-Validation Strategy and Grid Search

A single chronological split proved vulnerable to whichever atypical region fell in the validation window, so model selection used 3-fold chronological expanding-window cross-validation instead (Hespeler et al., 2025). Folds with no training attacks, or a single-class validation set, were excluded: a single-class set makes Average Precision (preferred over threshold-dependent F1) trivially 1.0 regardless of quality, confirmed in the pipeline's own output; such folds were skipped rather than assigned a fabricated score (Alvarez et al., 2022; Ok, Kluttermann and Muller, 2024). The winning configuration, by mean Average Precision across valid folds, was refit on the full training partition and tested once.



Figure 3.1: *Cross-validation log confirming degenerate single-class folds are skipped rather than scored.*

3.6.3 Class Imbalance Handling

Training attack rates ranged from 0.57% (BATADAL) to 5.83% (SWaT). Random Forest and CNN used SMOTE within each fold's training data only (Ahmed, Hameed and Bawany, 2022), ratios varying substantially (up to 15.7x for SWaT), an over-generalisation risk examined in Chapter 4 (Damtew and Chen, 2023). Isolation Forest's contamination parameter was set from the empirical attack rate; LSTM-AE trained exclusively on normal rows (Kravchik and Shabtai, 2022), needing no oversampling.

3.6.4 Windowing Parameters

CNN/LSTM-AE use fixed-length windows; Random Forest/Isolation Forest use unwindowed rows. SWaT/WaDi (1 Hz) use a 60-second window, 30-second stride; BATADAL (hourly) uses a 24-hour window, 12-hour stride. A window is labelled positive if any constituent row is, with a consequence explored in Chapter 4: a long attack collapses into few, similar windows under 50% overlap, compounding the SMOTE diversity concern above.

3.6.5 Model Architectures and Hyperparameter Search

Each model was tuned over a small, fixed grid (8 RF, 2 IF, 4 CNN, 2 LSTM-AE configurations): RF searched n_estimators, max_depth, min_samples_leaf; IF searched n_estimators; CNN searched filters/kernel_size; LSTM-AE searched latent_dim. Full search spaces and winning configurations per dataset/model are in Appendix A. No relationship was apparent between dataset size and configuration complexity, suggesting hyperparameter choice was a minor determinant relative to dataset-specific factors (Chapter 4).

3.6.6 Decision Threshold Selection

An absolute threshold set on validation data frequently failed to transfer to test, collapsing to predicting everything or nothing for several models despite genuine within-partition ranking ability (Duan et al., 2024; Leevy et al., 2023). This was resolved via a flag fraction: the F1-optimal fraction from validation is reapplied to test by recomputing the percentile cutoff fresh from test's own distribution, a claim about relative ranking rather than absolute comparability. Isolation Forest used its contamination parameter directly; a sensitivity grid (0.1-20% flag fractions) was also evaluated for transparency. Random seeds are fixed for Python, NumPy, and (random_state=42) Random Forest, Isolation Forest, and SMOTE throughout. TensorFlow determinism required setting TF_DETERMINISTIC_OPS and pinning thread counts to 1 before TensorFlow's own import, since it reads this configuration only at initialisation.

3.7 Evaluation Framework

Evaluation proceeds in three layers: classification accuracy (Section 3.7.1), operational resilience via detection and clearance latency (Section 3.7.2), and a composite Resilience-Accuracy Score (Section 3.7.3), applied per dataset (Section 3.7.4).

3.7.1 Accuracy Metrics

Precision, recall, and F1 are reported per dataset/model combination on the held-out test set at the validation-selected flag fraction. F1 also feeds the RAS composite (Section 3.7.3).

3.7.2 Resilience Metrics: MTTD and MTTR

Addressing Gap 2, two metrics are computed per attack event after segmenting each test partition. MTTD is elapsed time from an event's first row to its first correctly-flagged row; undetected events contribute to a separate miss rate rather than the average. MTTR departs from its conventional meaning, since this pipeline has no automated response: it is instead elapsed time from an event's end to the last trailing false-positive prediction, a proxy for alarm-fatigue burden (Halbouni et al., 2022), recorded as zero where none follow.

3.7.3 Resilience-Accuracy Score (RAS): AHP-Entropy Combined Weighting

RAS combines F1 (benefit) and MTTD/MTTR (cost criteria) via AHP-Entropy weighting (Zhou and Chen, 2025): subjective priority weighting (AHP) fused with objective, data-derived weighting (entropy). Each is min-max normalised within each dataset's four models:

$$\text{norm}(x_i) = (x_i - \min(x)) / (\max(x) - \min(x)) \quad (3.1)$$

$$\text{norm}(x_i) = (\max(x) - x_i) / (\max(x) - \min(x)) \quad (3.2)$$

Subjective weights derive from the pairwise AHP comparison matrix's principal eigenvector (Table 3.6, Saaty 1-9 scale):

$$w_{AHP} = \text{eigenvector}(A, \lambda_{max}) / \Sigma\, \text{eigenvector}(A, \lambda_{max}) \quad (3.3)$$

with CR = CI/RI (CI = $(\lambda\max - n)/(n - 1)$, RI Saaty's random-index constant, n=3). A Safety-Availability-Control priority (MTTD most critical, then F1, then MTTR) yields weights 0.320 (F1), 0.558 (MTTD), 0.122 (MTTR), CR=0.016, within Saaty's 0.1 threshold.

Objective entropy weights are computed per dataset from that dataset's own four normalised model scores:

$$p_{ij} = x_{ij} / \Sigma_i\, x_{ij}; \quad e_j = -(1/\ln n)\, \Sigma_i\, p_{ij} \ln(p_{ij}) \quad (3.4)$$

$$w_{entropy,j} = (1 - e_j) / \Sigma_j (1 - e_j) \quad (3.5)$$

where n=4 (models per dataset). MTTR carries zero entropy weight for WaDi specifically, since every WaDi model recorded MTTR of exactly zero, making the entropy weight mathematically zero regardless of MTTR's subjective importance. Weights are then combined by multiplication and renormalisation:

$$w_j = (w_{AHP,j} \times w_{entropy,j}) / \Sigma_j (w_{AHP,j} \times w_{entropy,j}) \quad (3.6)$$

giving each dataset's raw RAS as:

$$RAS_{raw} = \Sigma_j\, w_j \times \text{norm}_j \quad (3.7)$$

RAS is finally adjusted by a multiplicative miss-rate penalty rather than folding misses into MTTD, keeping MTTD interpretable as "how fast, when detection succeeds" while still capping models that miss most attacks:

$$RAS = RAS_{raw} \times (1 - \text{miss_rate}) \quad (3.8)$$

```
======================================================================
AHP subjective weights (from pairwise comparison matrix)
======================================================================
  F1    : 0.3196
  MTTD  : 0.5584
  MTTR  : 0.1220
  lambda_max=3.0183, CI=0.0091, CR=0.0158 (CONSISTENT, CR<0.1)

======================================================================
BATADAL
======================================================================
  Entropy weights (this dataset): F1=0.3815, MTTD=0.3107, MTTR=0.3078
  Combined weights:               F1=0.3662, MTTD=0.5211, MTTR=0.1127

  Model          F1  MTTD_n  MTTR_n  RAS_raw   Miss%  RAS_final
  RF          0.200   1.000   0.742   0.7510   57.1%     0.3222
  IF          0.067   0.792   1.000   0.5255   57.1%     0.2255
  CNN         0.400   0.623   0.000   0.6910   40.0%     0.4146
  LSTM-AE     0.222   0.000   0.667   0.2456   40.0%     0.1474

  RAS ranking: CNN > RF > IF > LSTM-AE
  Best by RAS: CNN  |  Best by raw F1: CNN  -> AGREES with F1 ranking

======================================================================
WaDi
======================================================================
  Entropy weights (this dataset): F1=0.4933, MTTD=0.5067, MTTR=0.0000
  Combined weights:               F1=0.3579, MTTD=0.6421, MTTR=0.0000

  Model          F1  MTTD_n  MTTR_n  RAS_raw   Miss%  RAS_final
  RF          0.075   1.000   1.000   0.7223   83.3%     0.1206
  IF          0.033   0.000   1.000   0.0353   66.7%     0.0117
  CNN         0.000   0.000   1.000   0.0000  100.0%     0.0000
  LSTM-AE     0.335   0.801   1.000   0.8725   33.3%     0.5819

  RAS ranking: LSTM-AE > RF > IF > CNN
  Best by RAS: LSTM-AE  |  Best by raw F1: LSTM-AE  -> AGREES with F1 ranking

======================================================================
SWaT
======================================================================
  Entropy weights (this dataset): F1=0.4187, MTTD=0.1935, MTTR=0.3878
  Combined weights:               F1=0.4627, MTTD=0.3737, MTTR=0.1636

  Model          F1  MTTD_n  MTTR_n  RAS_raw   Miss%  RAS_final
  RF          0.105   0.000   1.000   0.1636   57.1%     0.0702
  IF          0.326   0.970   0.916   0.9750   28.6%     0.6962
  CNN         0.112   1.000   0.000   0.3884   30.8%     0.2688
  LSTM-AE     0.294   0.974   0.062   0.7698   15.4%     0.6512

  RAS ranking: IF > LSTM-AE > CNN > RF
  Best by RAS: IF  |  Best by raw F1: IF  -> AGREES with F1 ranking
PS C:\Users\ajoku\AppData\Local\Programs\Microsoft VS Code>
```

Figure 3.2: *compute_ras.py output across all three datasets, showing AHP subjective weights, entropy weights, combined weights, and the full RAS computation (Tables 3.7-3.9).*

Tables 3.7-3.9 report the full computation per dataset.

Table 3.6: *AHP pairwise comparison matrix (Saaty 1–9 scale).*

Criterion	F1	MTTD	MTTR
F1	1	1/2	3
MTTD	2	1	4
MTTR	1/3	1/4	1

In WaDi and SWaT, the RAS ranking reproduces the raw F1 ranking across every model; BATADAL's exception is discussed in Section 4.4.

Table 3.7: *RAS computation, BATADAL (weights: F1=0.366, MTTD=0.521, MTTR=0.113).*

Model	F1	RAS (final)
RF	0.200	0.3222
IF	0.067	0.2255
CNN	0.400	0.4146
LSTM-AE	0.222	0.1474 (MTTD 100,800s)

Table 3.8: *RAS computation, WaDi (weights: F1=0.358, MTTD=0.642, MTTR=0.000).*

Model	F1	RAS (final)
RF	0.075	0.1206
IF	0.033	0.0117
CNN	0.000	0.0000
LSTM-AE	0.335	0.5819

Table 3.9: *RAS computation, SWaT (weights: F1=0.463, MTTD=0.374, MTTR=0.164).*

Model	F1	RAS (final)
RF	0.105	0.0702
IF	0.326	0.6962
CNN	0.112	0.2688
LSTM-AE	0.294	0.6512

3.7.4 Cross-Dataset Evaluation Protocol

RAS is computed independently within each dataset's four models, never pooled: MTTD/MTTR are absolute seconds, and BATADAL's hourly sampling and 24-hour

windowing places its latency on a fundamentally different scale (BATADAL LSTM-AE's 100,800s MTTD exceeds SWaT LSTM-AE's 21.8s by over 4,000x, from sampling resolution, not performance). This better operationalises RQ4, a within-deployment question; cross-dataset reliability (RQ2) is addressed qualitatively in Chapter 4 via relative ranking (Kim et al., 2023).

3.8 Explainable AI: SHAP Application and Quantitative Evaluation

Addressing Gap 1 and RQ3, SHAP was applied to all twelve trained combinations and evaluated against faithfulness, robustness, and complexity (Kalakoti et al., 2025; Nwakanma et al., 2023).

3.8.1 SHAP Explainer Selection and Implementation

Random Forest/Isolation Forest use TreeExplainer, computing exact Shapley values (for IF, its path-length anomaly score). CNN uses GradientExplainer on its sigmoid output. LSTM-AE, having no classification output, required a wrapper model (no added trainable parameters) outputting mean squared reconstruction error as a differentiable scalar, verified against direct computation, following precedent for autoencoder-based detectors (Muller et al., 2022).

```
============================================================
BATADAL / RF
============================================================
  Building TreeExplainer for RF...
  Computing SHAP values for 2000 instances...
/usr/local/lib/python3.13/dist-packages/sklearn/base.py:380: InconsistentVersionWarning: Trying to
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/sklearn/base.py:380: InconsistentVersionWarning: Trying to
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
    done in 19.7s
  Computing faithfulness (15 trials/instance)...
    done in 1640.0s
  Computing robustness (10 perturbations/instance)...
    done in 196.7s
{
  "faithfulness_mean": 0.9530522003300068,
  "faithfulness_nan_count": 0,
  "robustness_mean": 0.29046089286162274,
  "complexity_mean": 1.692066713913122,
  "n_instances": 2000
}

============================================================
BATADAL / IF
============================================================
  Building TreeExplainer for IF...
```

Figure 3.3: *SHAP explainer initialisation log confirming TreeExplainer and GradientExplainer per model.*

Table 3.10: *SHAP explainer selection by model.*

Model	Explainer	Explained quantity
RF	TreeExplainer	P(attack)
IF	TreeExplainer	Anomaly score
CNN	GradientExplainer	P(attack)
LSTM-AE	GradientExplainer (via wrapper)	Reconstruction error

3.8.2 Quantitative XAI Evaluation Metrics

Three metrics were implemented directly, since mixing scikit-learn and Keras precludes the single-framework Quantus toolkit (Hedstrom et al., 2023), per Kalakoti et al. (2025). Faithfulness is the Pearson correlation between attributed importance and

actual output change across randomly masked feature subsets. Robustness is the largest explanation change across small random perturbations (max-sensitivity principle); lower is more stable. Complexity is Shannon entropy of attribution values; lower indicates concentration on few features.

3.8.3 Sampling Methodology

Full-partition evaluation being intractable (up to 188,985 SWaT rows), stratified sampling was used: 2,000 instances (RF/IF) and up to 1,500 (CNN/LSTM-AE) per dataset (Kalakoti et al., 2025). BATADAL and WaDi's windowed partitions (214, 1,151 windows) fell below the 1,500 target and were evaluated in full; only SWaT (6,298) required subsampling. The SHAP background set (200 RF/IF, 100 windows CNN/LSTM-AE) is drawn from train only, avoiding leakage into the explanation baseline.

Table 3. 11: *Sampling summary by dataset.*

Dataset	RF / IF sample	CNN / LSTM-AE sample
BATADAL	2,000 of 2,588	214 (full population)
WaDi	2,000 of 34,561	1,151 (full population)
SWaT	2,000 of 188,985	1,500 of 6,298

3.8.4 Results

Table 3.12 covers all twelve combinations. Isolation Forest is the only model with negative faithfulness across all three datasets (-0.820, -0.908, -0.787), consistent with

its non-additive partitioning clashing with SHAP's additive assumption. LSTM-AE is the most stable model by far (0.005, 0.001, 0.022), likely because reconstruction error varies more smoothly than a decision boundary. SWaT CNN is the least stable combination (0.685), matching the probability saturation from Chapter 4: near-ceiling outputs leave little gradient signal, so small perturbations swing attributions disproportionately.

Table 3.12: *SHAP quantitative evaluation results, all combinations.*

Dataset	Model	Faithfulness	Robustness	Complexity
BATADAL	RF	0.953	0.291	1.692
BATADAL	IF	-0.820	0.243	2.352
BATADAL	CNN	0.565	0.052	2.328
BATADAL	LSTM-AE	0.186	0.005	2.164
WaDi	RF	-0.085	0.194	2.205
WaDi	IF	-0.908	0.145	2.640
WaDi	CNN	0.207	0.160	2.418
WaDi	LSTM-AE	0.465	0.001	2.453

SWaT	RF	0.466	0.107	2.478
SWaT	IF	-0.787	0.597	2.444
SWaT	CNN	0.928	0.685	1.340
SWaT	LSTM-AE	0.942	0.022	1.416

3.9 Cyber Threat Intelligence Integration

Addressing Gap 5 and RQ5, detections are mapped to MITRE ATT&CK for ICS techniques and enriched into a shareable format; Chapter 4's actionability framework is defined here. Per the scope in Section 1.6.2, this is a design-level demonstration: output is STIX-structured for every detection but not transmitted to a live TAXII server.

3.9.1 Detection-to-Technique Mapping Methodology

Mapping uses the SHAP explanations already computed (3.8): a flagged instance's most heavily-attributed tag identifies the implicated component (sensor, actuator, or downstream/derived), each mapped to candidate techniques under Impair Process Control (TA0106; MITRE, 2024): sensor suggests T0832 (Manipulation of View); actuator suggests T0855 (Unauthorized Command Message) or T0831/T0836 (Manipulation of Control/Modify Parameter); multi-tag patterns suggest T0827 (Loss of Control) or T0879 (Damage to Property). This is deliberately conservative, a triage aid rather than a certainty claim; distinguishing within a type needs corroborating evidence not captured here, left for future work.

Table 3.13: *SCADA tag type to candidate technique mapping.*

Tag type	Example tags	Candidate technique(s)
Sensor	LIT101, FIT101	T0832 Manipulation of View
Actuator	P102, MV302	T0855 Unauthorized Command Message; T0831 Manipulation of Control
Downstream / derived	Sustained cross-tag pattern	T0827 Loss of Control; T0879 Damage to Property

3.9.2 STIX-Structured Enrichment

Each detection is structured as a STIX Indicator object (OASIS, 2021) referencing its mapped technique. Custom x_-prefixed fields carry the detecting model's RAS score (Section 3.7.3) and its top SHAP feature. Illustrative structure:

{"type": "indicator", "pattern": "[scada-tag:name='MV302' AND scada-tag:value='unexpected-state']", "external_references": [{"source_name": "mitre-attack-ics", "external_id": "T0855"}], "x_ras_score": "<value>", "x_detecting_model": "<dataset>_<model>", "x_shap_top_feature": "MV302"}

3.9.3 Actionability Assessment Framework

RQ5 asks whether CTI integration improves actionability over standalone alerting. A framework compares a standalone alert against a CTI-enriched alert on four criteria: specificity, tactical context, component grounding, and shareability. A standalone alert

satisfies none by construction; a CTI-enriched alert satisfies all four. This does not measure actual analyst use, excluded from scope (1.6.2), but provides a reproducible comparison basis.

3.10 Ethical Considerations

This study uses only publicly available, non-personal OT/ICS testbed datasets (SWaT, WaDi, BATADAL) obtained through standard academic access routes, involving no primary data collection or human subjects. Formal ethics committee review was accordingly not required, consistent with this minimal-risk classification; no ethics approval reference number applies. The study is exclusively defensive, producing no capability usable against a live environment.

3.11 Summary of Methodology

Sections 3.2 to 3.10 together address all six research gaps and five research questions established in Chapters 1 and 2, from dataset provenance and forensic data-quality verification through model development, the RAS evaluation framework, SHAP explainability, and CTI integration. Chapter 4 turns to the results this methodology produced.

Chapter 4: Results

4.1 Introduction

This chapter reports and interprets the results produced by the methodology set out in Chapter 3, structured around the five research questions established in Section 1.5. Full result tables (3.7 to 3.9, 3.12) are referenced by number rather than reproduced.

4.2 RQ1 and RQ2: Detection Performance Across Models and Datasets

No single model dominates across all three datasets, directly addressing RQ1. Isolation Forest achieves the highest F1 on SWaT (0.326, Table 3.9) yet the lowest on BATADAL (0.067) and second lowest on WaDi (0.033, Tables 3.7 to 3.8), showing label-free training does not ensure consistent accuracy. LSTM-AE is the most consistent performer, ranking among the top two models on every dataset (WaDi 0.335, SWaT 0.294, BATADAL 0.222), suggesting its multivariate, time-dependent architecture transfers better across sampling resolutions (1 Hz for SWaT/WaDi, hourly for BATADAL) than the alternatives. Random Forest is comparatively flat (0.200, 0.075, 0.105), never best but never worst, consistent with prior findings that RF handles heterogeneous ICS data cheaply but without dataset-specific advantage (Wang et al., 2026; Hossain et al., 2024). CNN is the least stable model, strong on BATADAL (0.400) but failing outright on WaDi (0.000) and near the floor on SWaT (0.112), a divergence examined in Section 4.3. Together these results answer RQ2: supervised and unsupervised approaches each carry dataset-dependent strengths, with no

architecture reliably superior across OT contexts of differing structure and sampling rate.

4.3 SWaT CNN Saturation: Investigation and Interpretation

SWaT CNN's F1 of 0.112 conceals a measurement problem rather than simple low accuracy. Two hypotheses were tested. The first, that SMOTE amplified a narrow, low-diversity subset of attack windows drawn from the dense post-fusion block (Section 3.5.5), was ruled out: attack and normal window diversity proved statistically comparable.

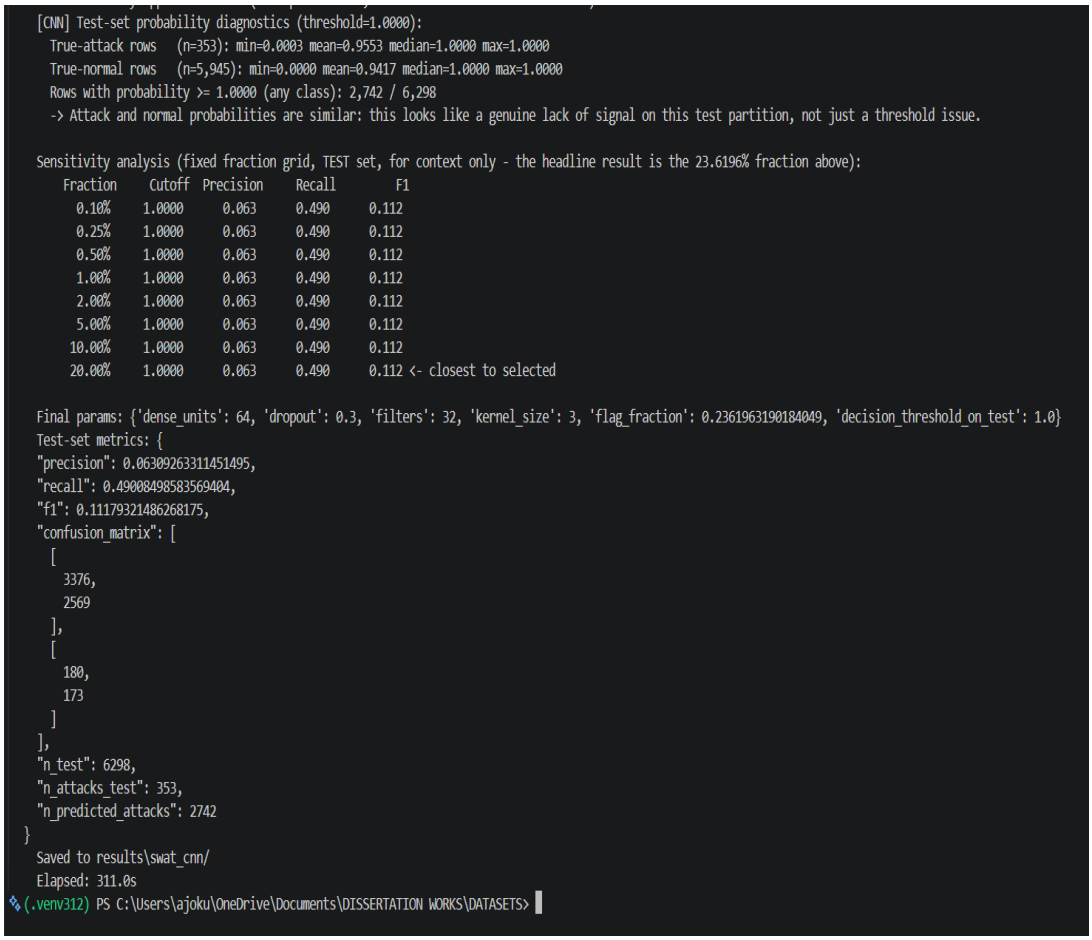


Figure 4.1: *SWaT CNN probability diagnostics, sensitivity analysis, and confusion matrix, confirming a decision threshold pinned at the probability ceiling (1.0) and F1 = 0.112 invariant across flag-fraction choice.*

The second, that FIT601's extreme values were compressed by MinMaxScaler normalisation, flattening the model's discriminative range, was also ruled out: FIT601's distribution proved a genuine bimodal on/off process signal, not rare-glitch outliers. The diagnostics confirmed predicted probabilities cluster near the classification ceiling for attack and normal instances alike, and the flag-fraction sensitivity curve (Section 3.6.6) remains flat across the full 0.1-20% range tested: the model has not learned a separable decision boundary, rather than merely being poorly thresholded. Set against WaDi CNN's F1 of 0.000, independently diagnosed as a genuine absence of learned signal, SWaT CNN represents a distinct failure mode of the same underlying weakness. Across two of three datasets the supervised deep architecture fails to separate classes in ways Random Forest and the unsupervised alternatives do not, a dataset-structure-dependent reliability gap directly relevant to RQ2 (Kravchik and Shabtai, 2022; Macas, Wu and Fuertes, 2022).

4.4 RQ4: Resilience-Accuracy Divergence

Tables 3.7 to 3.9 show RAS reproducing the F1 ranking exactly for WaDi and SWaT, but diverging for BATADAL: LSTM-AE ranks second by F1 (0.222) yet falls to last by RAS (0.147), driven by an MTTD of approximately 100,800 seconds, far exceeding the other three BATADAL models. This directly answers RQ4: the most accurate model is not consistently the most operationally resilient. Under BATADAL's hourly sampling and 24-hour windowing, LSTM-AE's slower detection carries an operational cost F1 alone cannot represent, supporting RAS's premise that dual evaluation captures what accuracy-only reporting omits (Alem et al., 2023; Lubis et al., 2025).

4.5 RQ3: SHAP Explainability and Operator Implications

Table 3.12 groups into three consistent patterns. Isolation Forest returns negative faithfulness on all three datasets, a structural mismatch between its interaction-driven anomaly scoring and SHAP's additive-decomposition assumption rather than an implementation defect, meaning its explanations should not be trusted operator-facing despite strong SWaT detection performance.

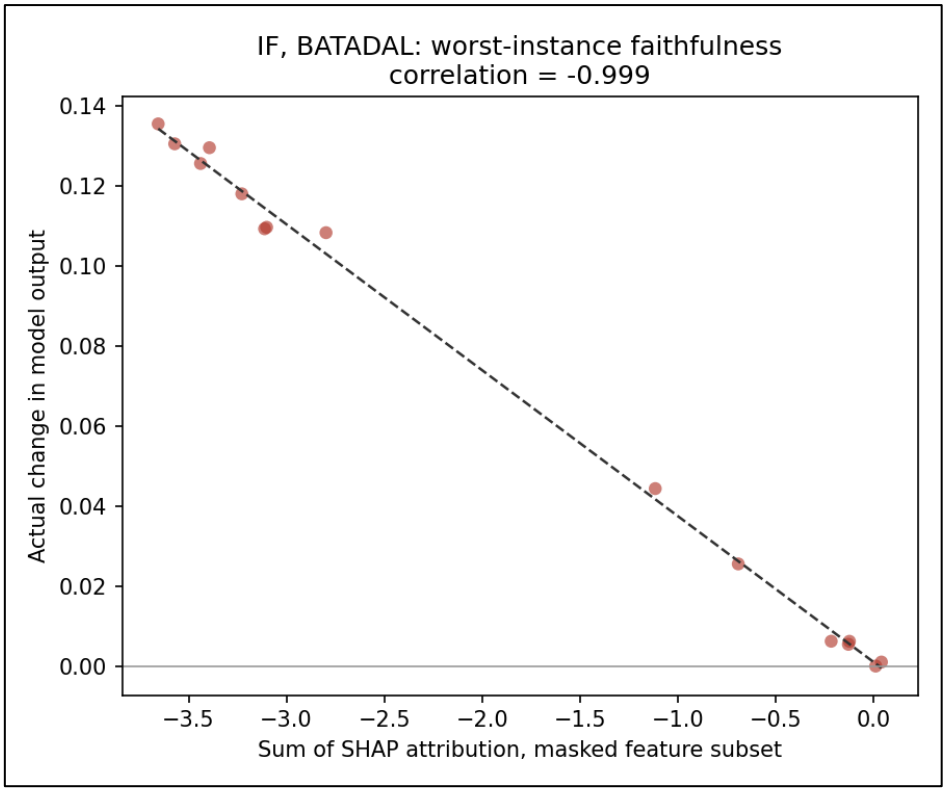


Figure 4.2: *SHAP faithfulness scatter, Isolation Forest, BATADAL, the single most negative instance evaluated (correlation = -0.999), illustrating the pattern behind the dataset-wide mean of -0.820 (Table 3.12).*

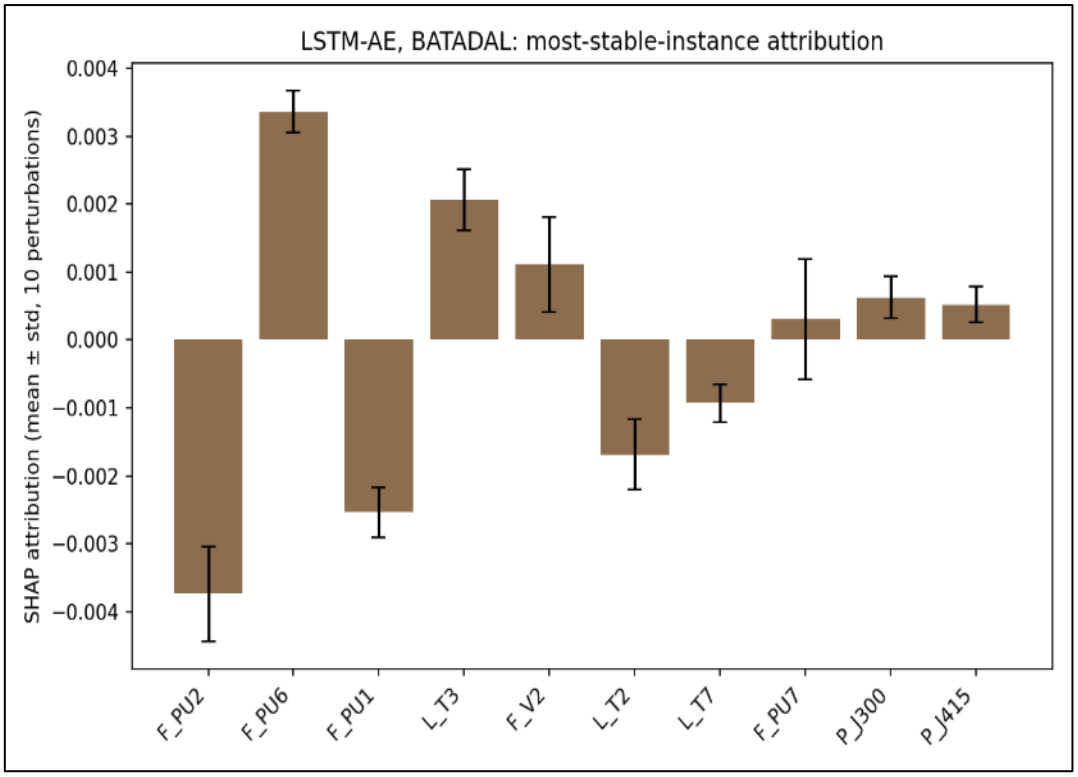


Figure 4.3: *SHAP attribution stability, LSTM Autoencoder, BATADAL, the single most stable instance evaluated (max change = 0.0026), illustrating the pattern behind the dataset-wide mean robustness of 0.0050 (Table 3.12).*

LSTM-AE is the most stable explainer by a wide margin on every dataset, suggesting reconstruction-error-based attributions are more trustworthy for operator review even at moderate F1. SWaT CNN's robustness score of 0.685, the least stable of any combination, independently corroborates Section 4.3's saturation finding: near-ceiling, non-separable probability outputs give little gradient signal, so small input perturbations produce disproportionate attribution swings (Kalakoti et al., 2025).

4.6 RQ5: CTI Actionability

Applying the Section 3.9.1 mapping to every instance evaluated in Section 3.8 produces 2,084 real STIX indicators across the three datasets (130 BATADAL, 202 WaDi, 1,752 SWaT), moving beyond a single worked example to an aggregate pattern.

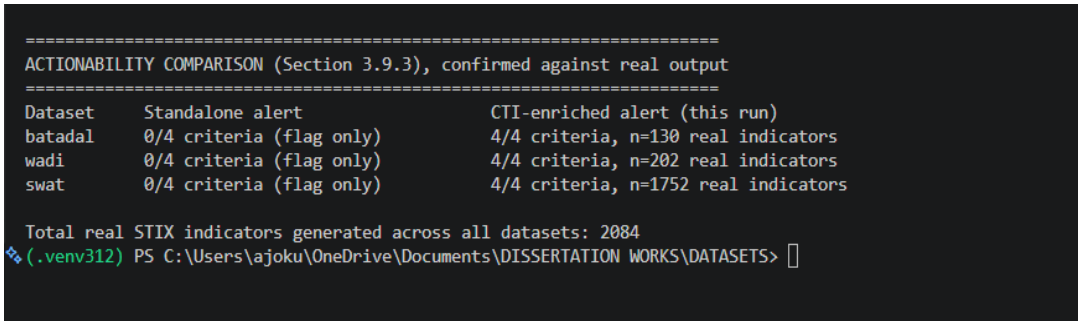


Figure 4.4: *mitre_attack_stix_mapping.py output confirming 2,084 real STIX indicators across all twelve combinations.*

The three datasets diverge sharply in implicated component type: BATADAL is actuator-dominant (82–100% of flagged instances per model), SWaT is sensor-dominant (87–100%), and WaDi is mixed, with CNN alone splitting 55% downstream, 38% actuator, 7% sensor. This pattern holds consistently across all four architectures within each dataset, suggesting a genuine structural difference in attack design between testbeds, not a modelling artefact.

One caveat: WaDi CNN's 58 enriched instances are not 58 real contextualised detections, since this model is already confirmed at F1 = 0.000 with inverted probability ranking (Section 4.2); they are most likely enriched false alarms.

Against the Section 3.9.3 framework, every standalone alert satisfies none of the four criteria by construction, while all 2,084 generated indicators satisfy all four. This answers RQ5 directly, CTI mapping measurably improves actionability at real scale, and surfaces a previously undocumented finding: the three testbeds' attacks differ structurally in the component type they target.

Chapter 5: Discussion

5.1 Introduction

This chapter interprets Chapter 4's findings against the literature reviewed in Chapter 2, examines CNN's dataset-dependent reliability as a broader generalisability finding, and closes the loop against each of the six research gaps identified in Section 1.3.

5.2 Detection Performance in Relation to the Literature

The absence of a single dominant model across SWaT, WaDi, and BATADAL (Section 4.2) qualifies claims of near-perfect cross-validated performance reported for autoencoder-plus-Random-Forest pipelines elsewhere (Hossain et al., 2024): such results appear attainable within a single dataset but do not generalise across this study's differing sampling rates and modalities. This is consistent with Wang et al.'s (2026) finding that traditional ensemble methods perform broadly equivalently across ICS benchmarks even under reduced training data, reflected here in Random Forest's flat, mid-range F1 across all three datasets. LSTM-AE's consistency, by contrast, supports Kravchik and Shabtai's (2022) argument for lightweight recurrent architectures over CNN-style feature extraction on multivariate ICS time-series, though Catillo et al.'s (2023) caution that reconstruction-error detection degrades when anomalies resemble normal patterns is directly borne out in Isolation Forest's volatility across datasets, less so in LSTM-AE's stability.

A further comparison: none of the XAI-for-ICS literature reviewed in Section 2.6, including Fung et al.'s (2024) systematic attribution-method comparison, tests SHAP against an isolation-based detector specifically. Isolation Forest's negative faithfulness (Section 3.8.4) is accordingly a finding this study contributes rather than confirms, and its generality beyond the three datasets tested here remains an open question rather than an established pattern.

5.3 CNN's Dataset-Dependent Reliability as a Generalisability Finding

CNN's split outcome, strong on BATADAL, saturated on SWaT, failed on WaDi, is this study's clearest generalisability evidence for RQ2. Rather than a single implementation defect, the pattern indicates that supervised deep architectures tuned to one OT structure (hourly, low-dimensional SCADA tags) do not straightforwardly transfer to another (high-frequency, high-dimensional sensor and actuator streams), cautioning against generalising CNN results from a single ICS dataset (Xi et al., 2024; Wang et al., 2024), consistent with calls for cross-dataset validation before claiming architecture superiority (Kim et al., 2023).

A more self-critical reading extends beyond architecture choice alone: WaDi's already low 5.77% attack rate is further thinned by the fixed 60-second, 50%-overlap windowing scheme applied uniformly across SWaT and WaDi (Section 3.6.4), a choice not varied by dataset. It remains possible that CNN's complete failure on WaDi (F1 = 0.000) reflects this windowing choice compounding label scarcity as much as it reflects the architecture itself, a confound the present single-windowing-scheme design cannot separate out.

5.4 Addressing the Six Research Gaps

Gap 1 (explainability) is addressed directly: SHAP was applied and quantitatively evaluated across all twelve combinations (Section 3.8, Table 3.12), moving beyond the feature-importance-only application Nwakanma et al. (2023) and Moustafa et al. (2023) identify as typical, surfacing Isolation Forest's negative faithfulness, a structural limitation a qualitative XAI application would not have revealed.

Gap 2 (resilience metrics) is addressed via MTTD and MTTR derived directly from model outputs and combined into RAS (Section 3.7), extending AlHidaifi, Asghar and Ansari's (2024) IT-focused resilience quantification into the ICS-specific operational terms Alem et al. (2023) identify as absent.

This extension is not without its own subjectivity: the Safety-Availability-Control priority ordering underlying the AHP weights (Section 3.7.3) reflects one defensible expert judgement among several plausible orderings, and a differently weighted judgement could shift BATADAL's LSTM-AE ranking further still, or partially recover it. RAS makes this trade-off explicit and auditable rather than eliminating it, a strength relative to an unweighted or purely data-driven alternative, though a genuine source of interpretive latitude.

Gap 3 (OT-specific context) is addressed through protocol-aware feature engineering on SWaT's genuine CIP/EtherNet-IP capture and physical-layer features across all three testbeds (Section 3.5), moving past the generic IT datasets Gaspar et al. (2024) note remain common.

Gap 4 (end-to-end pipeline) is addressed by the complete implementation from data ingestion through XAI and CTI output (Chapters 3 to 4), rather than the survey- or dataset-only contributions typifying much prior work (Duo et al., 2022; Halbouni et al., 2022).

Gap 5 (CTI integration) is addressed by the SHAP-to-MITRE-ATT&CK-for-ICS mapping and STIX-structured enrichment (Section 3.9), applied at real scale to 2,084 indicators (Section 4.6), closing the reactive-detection gap Hossain et al. (2024) and Pirca and Lallie (2023) describe, albeit at design level rather than live-feed integration, and further revealing an unanticipated structural difference between testbeds, BATADAL actuator-dominant, SWaT sensor-dominant, a byproduct of CTI integration beyond actionability alone.

Gap 6 (dual evaluation) is addressed most directly by RAS itself: Section 4.4's BATADAL finding, that the second most accurate model by F1 is the least resilient by RAS, is precisely the kind of result Alem et al. (2023) argue accuracy-only evaluation cannot surface, and goes beyond Morshedi and Matinkhah's (2025) Pareto-style accuracy-latency analysis by producing a single ranked composite score.

Whether a single composite score is preferable to a Pareto-front presentation is itself a design choice with a trade-off: RAS collapses two potentially non-commensurable dimensions into one ranking, which aids decision-making but obscures cases, such as WaDi, where the accuracy and resilience rankings already coincide and a Pareto view would have added little beyond what F1 alone showed.

Chapter 6: Conclusion

This research aimed to design, implement, and evaluate an AI-driven, explainable anomaly detection framework for ICS/OT environments, combining protocol-aware feature engineering, four supervised and unsupervised models, SHAP explainability, CTI enrichment, and a novel Resilience-Accuracy Score, across SWaT, WaDi, and BATADAL.

6.1 Research Questions Answered

RQ1 found no model uniformly superior at detection across datasets, with LSTM-AE the most consistent and CNN the least. RQ2 found detection performance dataset-structure-dependent rather than architecture-dependent, most visibly in CNN's contrasting BATADAL, SWaT, and WaDi outcomes. RQ3 found SHAP explanations informative but model-dependent in trustworthiness, with Isolation Forest's negative faithfulness a structural rather than incidental limitation. RQ4 found RAS diverges materially from F1 on BATADAL, confirming the most accurate model is not always the most operationally resilient. RQ5 found CTI enrichment measurably more actionable than standalone alerting, demonstrated across 2,084 real STIX indicators rather than only in the abstract, though live-feed transmission and automated response remain outside scope.

6.2 Contributions

This research's three principal contributions are the Resilience-Accuracy Score itself, an AHP-Entropy-weighted composite validated across three structurally distinct OT datasets and shown to surface findings F1 alone conceals; a fully quantitative SHAP evaluation, using faithfulness, robustness, and complexity rather than qualitative feature-importance inspection, across all twelve trained combinations; and a working SHAP-to-MITRE-ATT&CK-for-ICS-to-STIX mapping demonstrating a concrete path from model output to shareable threat intelligence.

6.3 Limitations

Several limitations qualify these contributions. WaDi's post-gap timestamp reconstruction (Appendix A.2.8) rests on a right-alignment assumption that, while the best-supported reading of the available evidence, is not independently verified. SWaT's physical ground-truth label carries an unresolved 721-second block (Section 3.5.5, Appendix A.1.8 to A.1.9) where the available diagnostic evidence neither confirms nor excludes a genuine undocumented event. SWaT CNN's saturation, though investigated via two ruled-out hypotheses (Section 4.3), remains an open question as to root cause rather than a fully explained mechanism. CTI integration was demonstrated at design level only, without live TAXII transmission, automated response, or a formal operator user study (Section 1.6.2). Finally, an ordering error in the TensorFlow determinism configuration meant this seeding was not yet in effect when BATADAL's CNN and LSTM-AE models were trained; those two results are genuine outputs of an actual run

but are not guaranteed to reproduce bit-for-bit on re-execution, unlike every other combination, trained after the error was corrected.

6.4 Future Work

Future work should prioritise a mechanistic explanation for SWaT CNN's saturation beyond the two hypotheses ruled out here, live CTI feed integration with automated or semi-automated response, and formal operator studies validating the actionability claims in Section 4.6 against real analyst behaviour rather than a structural framework alone.

This research demonstrates that combining accuracy-oriented and resilience-oriented evaluation, grounded in operator-facing explainability and threat-intelligence enrichment, surfaces operationally material findings that detection accuracy alone would leave hidden, directly answering the aim with which this study began.

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Appendix A: Full Implementation and Diagnostic Detail

This appendix provides the full implementation and forensic diagnostic detail underlying the summarised findings reported in Sections 3.5 and 3.6. It closes the Appendix A references made throughout those sections and preserves the complete investigative trail behind each data-quality finding at the level of detail it was originally conducted, rather than the condensed outcome-only account given in the main chapter.

A.1 SWaT Network-Physical Fusion: Full Diagnostic Trail

A.1.1 Pipeline implementation and checkpointing

The 784 parsed network log files supplied by iTrust are organised into thirteen date-labelled folders spanning 21/22 December 2015 to 3 January 2016. A batch-processing script (LOAD_NETWORK.py) was developed to: (i) load and concatenate all raw files within a single date folder; (ii) aggregate the concatenated packets into one row per second per SCADA tag, comprising request/response counts, read-write ratios, and CIP transaction-identifier sequencing checks (corrected for the counter's observed wraparound behaviour at 65,536); (iii) write each date folder's aggregated output to an independent checkpoint file before proceeding to the next folder, so a failure partway through the multi-hour batch run would not require reprocessing already-completed folders; and (iv) fuse the concatenation of all thirteen checkpoints against the combined SWaT physical historian timeline via a nearest-timestamp join with a one-second tolerance, producing the fused table of 1,055,475 rows.

A.1.2 Correction of a corrupted timestamp field

A diagnostic script compared each date folder's aggregated minimum and maximum timestamp against the expected campaign window (20 December 2015 to 4 January 2016). This flagged a single row bearing the date 5 December 2015, twenty days outside the containing checkpoint file's otherwise correct range. A targeted scan of the corresponding raw network files traced the row to 2015-12-25_151922_85.log.part05_sorted.csv. The row's time field (11:35:41) was identical to that of multiple immediately surrounding rows correctly dated 25 December 2015, within the same hour, minute, and second; this uniformity supported the conclusion that the date field alone had been corrupted at source. The date field was corrected from 5 December 2015 to 25 December 2015 directly in the raw file, and the affected checkpoint was regenerated.

A.1.3 Resolution of duplicate boundary-second records

Independent per-date aggregation produced twelve duplicated timestamps at the boundaries between consecutive date folders, each corresponding to a second whose constituent packets had been split across two folders by the raw capture's file-partitioning. Rather than discarding one copy, which would understate the true packet count for that second, duplicate rows were merged by summing all count-based features and recomputing derived ratios from the summed totals, rather than averaging two already-partial ratios.

A.1.4 Dual ground-truth agreement assessment

The fused output carries two independently derived attack labels for each second: the physical historian's own Normal/Attack field, and the network layer's embedded per-

second attack flag, aggregated as the maximum value across all SCADA tags polled within that second. Comparing the two labels across the full 1,055,475-second fused output yielded a raw agreement rate of 99.91 per cent, with 922 seconds of disagreement.

A.1.5 Classification of disagreements against documented attack windows

To establish whether the 922 disagreements related to genuine attacks, the temporal distance of each disagreeing second to the nearest of the 36 documented attack windows (List_of_attacks_Final) was computed. Prior to this comparison, a transcription error in the source document was identified and corrected: the final five documented attacks were dated 02/01/2015, inconsistent with the campaign's actual December 2015 to January 2016 span and with the immediately preceding, correctly dated 01/01/2016 entries; these were corrected to 02/01/2016. Classification against the corrected attack windows found that 92 disagreements (9.98 per cent) fell inside a documented attack window, 20 (2.17 per cent) fell within five seconds of one, and a further 88 fell between six and sixty seconds of a window boundary. Combined, 200 of the 922 disagreements (21.7 per cent) were attributable to short cross-layer detection lag at attack boundaries. The remaining 722 (78.3 per cent) were more than sixty minutes from any documented attack, indicating a distinct, unrelated population rather than a continuum of decreasing proximity.

A.1.6 Structural analysis of the unexplained disagreements

The 722 distant disagreements were further analysed by disagreement direction, hour-of-day distribution, calendar date, and run-length structure. This analysis established that the 722 seconds were not independently scattered but comprised exactly two

components: one isolated second, and a single unbroken run of 721 consecutive seconds. Throughout this 721-second run (29 December 2015, 06:30:00 to 06:42:00), every second showed the physical label as Attack and the network label as Normal, and no discrete actuator or valve changed state. The nearest documented attack to this window began approximately four hours and forty-one minutes later, ruling out a simple date or time transcription error as the explanation. Three candidate explanations were identified for further investigation: a genuine undocumented attack with a network-layer detection gap; an artefact of a data-acquisition or historian communication interruption during that period; or a labelling defect confined to that stretch of the physical file's own ground-truth column.

A.1.7 Resolution of the isolated single-second anomaly

The one remaining isolated second, timestamped 25 December 2015, 16:24:40 and distinct from the 721-second block above, was traced via a targeted raw-packet scan to 2015-12-25_203211_86.log.part04_sorted.csv. Inspection identified that a comma had replaced the final period of a destination IP address field within a single row (192.168.1.20 recorded as two separate fields, 192.168 and 1.2), shifting every subsequent field in that row one position to the right and placing the true source-port value (53250) in the position expected for the network-layer attack flag. Correcting this single delimiter restored the row's intended flag value of 0, consistent with every neighbouring row at that timestamp, and removed this second from the disagreement count, reducing the total from 922 to 921.

A.1.8 Investigation of the residual 721-second block

The physical plausibility of the remaining 721-second window was assessed in stages. An initial comparison of window-period sensor means against a pre-window baseline flagged two analog tags, AIT503 and LIT101, as significantly shifted (a third apparent shift was discounted as an artefact of a near-zero baseline standard deviation); no discrete actuator changed state throughout the window. Because a mean-shift comparison alone cannot detect a sensor reading that has become unnaturally static, a variance-based analysis was subsequently undertaken across all 25 analog process variables, comparing during-window variance against a three-hour surrounding baseline. This flagged four flow and differential-pressure tags (FIT601, FIT301, DPIT301, FIT201) as showing markedly reduced variance, and separately flagged LIT101 as apparently frozen.

Because tank-level tags such as LIT101 were observed, over the course of this investigation, to exhibit slow swings across multi-hour periods, a raw-variance comparison against a long baseline was recognised as liable to misclassify any short window drawn from such a cycle as artificially quiet. The analysis was therefore repeated using the second-to-second first difference of each signal, which removes slow trend and cyclical components while preserving genuine local sensor noise (Box et al., 2015), together with an explicit calculation of the longest continuous run of identical consecutive values. This corrected analysis confirmed that LIT101, together with the related tank-level tags LIT301 and LIT401, was not genuinely anomalous during the window, its reduced short-window variance being fully attributable to its position within an ongoing, unrelated multi-hour trend. Of the four flow and pressure tags, only

FIT601 exhibited a sustained held value, persisting for 278 of the 721 seconds (38.6 per cent of the window) and beginning at precisely 06:30:00, the exact onset of the physical label transition. The remaining three tags showed only brief, non-sustained periods of reduced variability, ranging from fifteen to twenty-nine seconds.

A.1.9 Treatment of the residual anomaly

Because FIT601's held-value period does not span the full labelled window, and no other tag exhibited a comparably sustained pattern, the available evidence does not conclusively support either a coordinated attack-concealment mechanism or a full data-acquisition interruption as the explanation for this window. The evidence most plausibly supports either a genuine but undocumented event affecting one sensor circuit, or a labelling defect isolated to this stretch of the physical file; a full data-link interruption is not well supported, since such an interruption would be expected to affect all instrumented channels for the full duration of the window, which was not observed. Accordingly, the 721-second block is documented as an investigated, partially-evidenced but unresolved data-quality characteristic of the SWaT Attack_v0 physical label file, rather than confirmed as an additional attack instance.

A.1.10 Discovery and correction of a network-physical fusion boundary gap

Unlike the defects documented in A.1.2 to A.1.9, which were surfaced during the initial forensic verification of the fused output, this defect was identified only once model training began (Section 3.6): a substantial share of the physical-layer feature columns returned missing values, at a rate high enough to warrant investigation. The original verification pass had checked timestamp integrity, duplicate detection, and cross-layer label agreement, none of which would surface a completeness gap in the feature values

themselves: a row can carry a valid, non-duplicate timestamp and a correctly agreeing label while every physical-sensor column beneath it is empty.

A targeted diagnostic (diagnose_swat_nan.py) found that all 51 physical-layer columns shared an identical missingness rate for every affected row, strong evidence that the gap originated at the row level, from the fusion join itself, rather than from any individual sensor. A follow-up temporal analysis (diagnose_swat_nan_timeline.py) confirmed the missing rows formed three contiguous blocks: one of 65,523 rows beginning at the campaign's literal first timestamp (21 December 2015, 22:17:56) and running for approximately 18 hours 12 minutes; one of 44,950 rows running from 2 January 2016, 15:00:01 through to the campaign's literal last timestamp (3 January 2016, 03:29:10), approximately 12 hours 29 minutes; and one isolated 79-second block beginning 31 December 2015, 21:00:38.

The two large blocks sit precisely at the start and end of the network capture, indicating the physical historian began recording approximately 18 hours after the network capture started, and stopped recording approximately 12.5 hours before it ended. For these windows, the fusion join correctly finds no physical-historian reading within its one-second tolerance, because none exists: a genuine absence of data at the collection boundary, not a corrupted or lost value. These 110,552 rows (10.47 per cent of the fused output) are dropped entirely rather than imputed, in data_loading.py::load_swat(), prior to the chronological split. Imputation was rejected on the same grounds established for this pipeline: fabricating physical readings for a period in which the historian was not yet operating would not recover lost information, it would manufacture information that never existed. The split is recomputed on the resulting, complete-data-only 944,923-

row timeline, giving a training partition of 755,938 rows (44,086 attacks, 5.83 per cent) and test partition of 188,985 rows (9,763 attacks, 5.17 per cent).

A.2 WaDi and BATADAL: Full Diagnostic Trail

A.2.1 Diagnostic script design

A single diagnostic script (Wadi & Batadal Check.py) was developed to mirror the checks already applied to SWaT: missing-value counts per column, duplicate-timestamp detection, date-range verification against each file's documented collection window, attack-label distribution, and schema alignment between each dataset's normal and attack files. Because column names differ across files and are not always known in advance, a case-insensitive helper function (find_column) was used to locate date, time, and label columns by substring match, returning a warning rather than raising an exception where a column could not be located.

A.2.2 BATADAL: verification pass

Running the diagnostic against BATADAL_dataset03_Baseline.csv (8,761 rows, 45 columns) found no missing values in any column, zero duplicate timestamps, a date range of 6 January 2014 to 6 January 2015 exactly matching the documented collection window, and an ATT_FLAG value of 0 across all rows. Running the same diagnostic against BATADAL_dataset04_Attack.csv (4,177 rows, 45 columns) likewise found no missing values and zero duplicate timestamps, with a date range of 4 July 2016 to 25 December 2016 matching expectations. ATT_FLAG values of -999 (3,958 rows) and 1 (219 rows) were observed, with no rows explicitly coded 0; the 219 rows flagged 1 correspond exactly to the cumulative labelled-hour total of the seven documented

BATADAL attacks, confirming the label column is intact and correctly aligned. A column-naming defect was subsequently identified: the ATT_FLAG column header in BATADAL_dataset04_Attack.csv carried a leading whitespace character, absent from the corresponding column in BATADAL_dataset03_Baseline.csv. Naive concatenation of the two files therefore produced two nominally identical but programmatically distinct columns, silently excluding all 219 attack-labelled rows from downstream analysis rather than raising an error. The defect was traced by comparing the post-concatenation attack-row count (0) against the documented attack-hour total of 219, and corrected by removing the leading whitespace from the source file's header row.

A.2.3 WaDi: initial diagnostic and identification of two distinct defects

Running the same diagnostic against WADI_14days_new.csv (784,571 rows, 130 columns) initially returned a date-parse failure rate of 82.78 per cent and a correspondingly implausible duplicate-timestamp count, alongside a small number of physically-plausible missing values. Running it against WADI_attackdataLABLE.csv (172,804 rows, 131 columns) returned no identifiable date, time, or label columns at all, with every column instead named as a bare integer. Subsequent investigation traced these outputs to two unrelated defects: a structural header offset confined to the attack file, and a corrupted Time field present in both files.

A.2.4 Resolution of the attack file's header offset

Raw inspection of the first lines of WADI_attackdataLABLE.csv showed that line 0 is a stray row of bare column indices, with the genuine header, the same tag names used in the normal file plus the label column, appearing on line 1. Every column name

reported by the diagnostic as an integer was therefore an artefact of pandas reading line 0 as the header rather than a genuine schema defect. The fix applied was to skip the first line when reading the file (skiprows=1). Re-running the diagnostic against the corrected read recovered all 131 named columns, including the label column Attack LABLE (1: No Attack, -1: Attack).

A.2.5 Diagnosis of the corrupted Time field

With the header issue resolved, raw Date/Time values were sampled at points spread across both files. Every sampled value, in both files, was found to be a fixed seven-character string of the form MM:SS.f (for example 16:40.0, 33:20.0, 46:40.0), with no hour component present at all. Minute values exceeding 59 confirmed this is not a conventional clock reading but a continuous elapsed-seconds counter that wraps every 3,600 seconds, with the hour digit dropped entirely at the point of export. This explains the original 82.78 per cent parse-failure rate: naive parsing of the combined Date and Time strings failed for the majority of rows because the Time component alone cannot be resolved to an absolute time of day, and, because pandas treats unparsed (NaT) values as mutually equal under duplicate-detection, the resulting duplicate-timestamp count was correspondingly inflated rather than reflecting genuine duplication in the source data.

A.2.6 Timestamp reconstruction methodology

Because the Time field cannot supply reliable hour information, it was discarded as a basis for reconstruction. Both files were instead assumed to reflect continuous 1 Hz sampling within each calendar-date group, consistent with iTrust's documented WaDi sampling rate, with each row's reconstructed timestamp built from the parsed Date plus

its row position within that date group. A further complication emerged during this step: the raw Date field mixes a four-digit-year representation (for example 9/25/2017) with a two-digit-year representation (for example 10/2/17), with the transition occurring partway through each file. An initial reconstruction attempt using a single fixed format left the majority of affected rows unparsed (649,452 of 784,571 in the normal file). Parsing was corrected to attempt both formats in sequence, taking the first successful result for each row, which resolved all remaining unparsed dates in both files.

A.2.7 Discovery and confirmation of a data-acquisition gap in the WaDi normal file

With dual-format Date parsing in place, the mm:ss consistency check on the normal file fell from 3.37 per cent to near-zero everywhere except a single boundary, between the last row dated 29 September 2017 and the first row dated 2 October 2017. No rows dated 30 September or 1 October were present in the file at all. Two independent signals confirmed this was a genuine acquisition gap rather than a labelling artefact. First, the file's own raw Row index, an untouched sequential field independent of both Date and Time, jumps from 335,999 to 600,000 at exactly this boundary, a discontinuity of 264,001 rows, equivalent to approximately 73.3 hours of missing samples at the file's otherwise-continuous 1 Hz rate. Second, the raw Time field itself does not reset at this boundary (the sequence runs 39:59.0 directly into 40:00.0, as though no interruption occurred), confirming it is a free-running counter blind to calendar-date changes.

A.2.8 Right-alignment of the post-gap partial day (2 October 2017)

A straightforward reconstruction, resetting the elapsed-seconds counter to zero at the start of every calendar date, would place the 26,401 rows carrying the 2 October 2017 date label at the beginning of that day, spanning 00:00:00 to 07:20:00. This placement

was rejected in favour of right-alignment: the immediately following date, 3 October 2017, contains the full complement of 86,400 rows expected of an uninterrupted day and validates cleanly under the simple midnight-reset assumption, and no further Row-index discontinuity was detected between the end of 2 October's rows and the start of 3 October. Since only one gap was detected across the whole file, and it lies before 2 October's rows rather than within them, the balance of evidence supports treating those 26,401 rows as the tail end of continuous post-gap logging running straight through into the independently validated start of 3 October, rather than as a fresh restart at 2 October's own midnight. This right-alignment is reported as the best-supported reconstruction available from the evidence, not as an independently verified fact, and is noted as a limitation in Chapter 6.

Table A.1: *Progression of the WaDi normal-operation file's timestamp reconstruction.*

Diagnostic stage	Finding	Rows affected
Single-format Date parse (initial)	Majority of rows unparsed; downstream counts inflated as an NaT artefact, not a genuine defect	649,452 of 784,571 (82.78%)
Dual-format Date parse (corrected)	All rows parsed under one of the two formats present	0 unparsed
mm:ss vs original Time field check	Disagreement concentrated entirely at the 29 Sept / 2 Oct boundary	26,401 (3.37%)

Raw Row-index inspection	Discontinuity from row 335,999 to 600,000, confirming a genuine acquisition gap	264,001-row jump (~73.3 hours at 1 Hz)
Right-alignment of 2 Oct	Reconstructed as 16:39:59-23:59:59, immediately preceding the validated start of 3 Oct	26,401
Final reconstructed file	Range 25 Sept 2017 00:00:00 to 7 Oct 2017 21:16:10; zero duplicate timestamps	784,571

A.2.9 WaDi attack file: reconstruction and validation

The same dual-format Date parsing and 1 Hz row-position reconstruction was applied to WADI_attackdataLABLE.csv following the header correction, achieving zero unparsed dates, a 0 per cent mm:ss consistency mismatch, and zero duplicate timestamps across all rows carrying a valid date. The reconstructed range spans 9 October 2017 00:00:00 to 11 October 2017 18:00:00, matching iTrust's documented attack-collection window. The interior-partial-day test used to detect the normal file's gap found no such case in the attack file: the only date groups with fewer than 86,400 rows are the first (9 October, 21,600 rows) and last (11 October, 64,801 rows), the expected start and end boundaries of a three-day collection window, and no right-alignment correction was applied. Two rows at the very end of the file (index 172,801 and 172,802) carried entirely null Date, Time, and feature values, identified as a stray

end-of-file artefact and dropped. Across the 172,801 valid rows, the label column Attack LABLE records 162,824 rows labelled 1 (no attack) and 9,977 rows labelled -1 (attack), a 5.77 per cent attack rate.

A.2.10 Summary

Table A.2: *Implementation and data-quality outcomes, WaDi and BATADAL physical/process-layer files.*

File	Rows	Defects identified	Status
BATADAL dataset03 (baseline)	8,761	None	Verified clean; no correction required
BATADAL dataset04 (attack)	4,177	Column-naming defect (leading whitespace in ATT_FLAG header)	Corrected at source; ATT_FLAG matches the 219 documented attack hours
WaDi normal (WADI_14days_new)	784,571	Corrupted Time field; mixed Date formats; ~3-day acquisition gap (30 Sept-1 Oct 2017)	Reconstructed and validated; post-gap right-alignment

			documented as a stated assumption
WaDi attack (WADI_attackdataLABLE)	172,801	Header offset; mixed Date formats; 2 trailing blank rows	Reconstructed and validated; no acquisition gap detected

A.3 Hyperparameter Search Spaces and Grid-Search Winners

Each of the four architectures introduced in Section 3.6.1 was tuned over a small, fixed grid of hyperparameters, kept deliberately modest given the 3-fold cross-validation multiplier already applied to each candidate (Section 3.6.2) and the correspondingly larger computational cost of a wider search.

Table A.3: *Hyperparameter search space by model.*

Model	Hyperparameters searched	Values	Fixed parameters
Random Forest	n_estimators, max_depth, min_samples_leaf	{100, 200} x {None, 20} x {1, 5}	N/A

Isolation Forest	n_estimators	{100, 200}	contamination (set from data, 3.6.3)
CNN	filters, kernel_size	{32, 64} x {3, 5}	dense_units=64, dropout=0.3
LSTM-AE	latent_dim	{16, 32}	dropout=0.2

This gives eight candidate configurations for Random Forest, two for Isolation Forest, four for CNN, and two for LSTM Autoencoder, evaluated per dataset via the cross-validation procedure described in Section 3.6.2. Table A.4 reports the winning configuration selected for each dataset/model combination, together with the decision threshold ultimately selected for that configuration.

Table A.4: *Grid-search winners and selected operating point, per dataset and model.*

Dataset	Model	Selected hyperparameters	Flag fraction / contamination
BATADAL	RF	max_depth=None, min_samples_leaf=5, n_estimators=200	4.24%
BATADAL	IF	n_estimators=100	contamination=0.0057
BATADAL	CNN	filters=32, kernel_size=5	7.69%

BATADAL	LSTM-AE	latent_dim=32	7.69%
WaDi	RF	max_depth=20, min_samples_leaf=1, n_estimators=100	3.60%
WaDi	IF	n_estimators=200	contamination=0.0074
WaDi	CNN	filters=32, kernel_size=3	5.02%
WaDi	LSTM-AE	latent_dim=32	4.26%
SWaT	RF	max_depth=None, min_samples_leaf=5, n_estimators=100	27.44%
SWaT	IF	n_estimators=200	contamination=0.0583
SWaT	CNN	filters=32, kernel_size=3	23.62%
SWaT	LSTM-AE	latent_dim=16	21.93%

No systematic relationship is apparent between dataset size and the winning configuration's complexity: the smaller, shallower configurations win on some datasets and the larger ones on others, suggesting that, within the modest range searched, hyperparameter choice was a comparatively minor determinant of outcome relative to the dataset-specific factors examined in Chapter 4.

MSc Project Plan

CSC-44120

Project Overview and Description

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Degree Title:
MSc Cyber Security
Supervisor Name:
Dr Mohamed Ghanem
Project Title:
AI-Enhanced Network Traffic and Process Analytics for Threat Detection and Cyber Resilience Metric Optimisation (OT/ICS Focus): A SWaT-Centred Study with Cross-Dataset Generalisability Testing
Please provide a brief Project Description:

This project develops an AI-enhanced network traffic analytics framework specifically targeting Operational Technology (OT) and Industrial Control Systems (ICS) environments. Existing intrusion detection systems predominantly address IT-domain threats and fail to account for the Safety, Availability, and Control (SAC) priorities of industrial environments. The project addresses this gap by building an end-to-end ML pipeline that ingests OT network traffic, performs protocol-aware feature engineering, detects anomalies using ensemble and deep learning classifiers, and outputs a novel Resilience-Accuracy Score (RAS) that combines detection accuracy (F1) with operational resilience metrics (Mean Time to Detect and Mean Time to Respond). Explainable AI (XAI) via SHAP is integrated to provide operator-facing transparency. The system is validated against OT-specific datasets including SWaT, WaDi and BATADAL.

What are the aims and objectives of the Project?

Aim:
To design, implement, and evaluate an AI-driven network traffic analytics system for OT/ICS environments that improves both threat detection accuracy and cyber resilience, measured through the novel Resilience-Accuracy Score (RAS).

Objectives:

1. Conduct a systematic literature review of ML-based anomaly detection in

OT/ICS environments, identifying six key research gaps.
2. Design and implement an OT-specific AI pipeline incorporating protocol-aware feature engineering for Modbus, DNP3, and OPC UA traffic.
3. Train, tune, and evaluate multiple ML models (Random Forest, Isolation Forest, LSTM, CNN) on OT datasets (SWaT, WaDi, BATADAL).
4. Define and compute the Resilience-Accuracy Score (RAS) using AHP-Entropy combined weighting of F1, normalised MTTD, and normalised MTTR.
5. Integrate SHAP-based Explainable AI to produce operator-facing feature attribution explanations for anomaly detections.
6. Evaluate the system against the six identified gaps and validate the RAS as a novel dual-metric contribution.

Please provide a brief overview of the key literature related to the Project:

The literature review covers 27 verified comparable works spanning three themes:

i. AI/ML-based anomaly detection in OT/ICS (Mishra et al., 2025; Nguyen et al., 2024; Oyedotun et al., 2025; Benka et al., 2025; Mubarak et al., 2022);
ii. cyber resilience quantification (AlHidaifi et al., 2024; Kotliarov & Bortnik, 2025; Presekal et al., 2023); and
iii. explainable and hybrid IDS for SCADA/ICS (Skrodelis & Romanovs, 2026; De Paola et al., 2024; Sun et al., 2023).

Six persistent research gaps were identified across all 27 works: absence of MTTD/MTTR-based resilience metrics; lack of XAI for OT operators; exclusive use of IT/IoT datasets; survey/dataset-only contributions; no Cyber Threat Intelligence integration; and no dual evaluation combining accuracy and resilience. No prior work addresses all six gaps simultaneously. The RAS is proposed as a direct response to Gap 6, which applies universally across all 27 comparable works.

Project Processes and Methods

Please provide a brief overview of the Methodology to be used in the Project (inc. an overview of best practice within the Methodology):

The project adopts a Design Science Research (DSR) methodology, constructing and evaluating an artefact (the AI-enhanced OT IDS pipeline) against measurable performance criteria. The methodology comprises five phases:

Phase 1 (Data Acquisition and Preprocessing): OT/ICS-specific datasets (SWaT, WaDi, BATADAL) are loaded, cleaned, and normalised. Protocol-aware features specific to Modbus, DNP3, and OPC UA are extracted alongside standard flow features.

Phase 2 (Feature Engineering and Selection): Three-stage feature selection using correlation filtering, mutual information scoring, and SHAP-guided importance ranking to identify the most discriminating OT traffic features.

Phase 3 (Model Training and Evaluation): Supervised (Random Forest, CNN) and unsupervised (Isolation Forest, LSTM Autoencoder) models are trained, tuned via grid search, and evaluated on F1, precision, recall, MTTD, and MTTR.
Phase 4 (RAS Computation): The Resilience-Accuracy Score is computed using AHP-Entropy combined weighting. Sensitivity analysis validates weight robustness.
Phase 5 (XAI Integration and Evaluation): SHAP TreeExplainer generates feature attribution explanations. Results are evaluated against the six identified research gaps to demonstrate original contribution.
Best practice alignment: CRISP-DM for ML pipeline structure; NIST SP 800-82 for OT security context; IEC 62443 for zone/conduit threat modelling; MITRE ATT&CK for ICS for attack scenario mapping.

Will any special Data Collection Methods will be employed (e.g card sorts, questionnaires, simulations, …)?

No primary human data collection is required. The project uses three publicly available OT/ICS benchmark datasets:

1. **SWaT (Secure Water Treatment) dataset:** iTrust, Singapore University of Technology and Design. Six-stage water treatment testbed; seven days of normal operation and a six-day attack phase with 36 documented attacks. The only dataset providing genuine network-layer traffic alongside physical process data (Goh et al., 2017).
2. **WaDi (Water Distribution) dataset:** iTrust, Singapore University of Technology and Design. Water distribution network downstream of SWaT; 14 days of normal operation and 2 days of attack scenarios across 123 sensors and actuators. Contributes physical-layer data only, since its 2023 network capture is temporally disjoint from its 2017 attack labels (Ahmed, Palleti and Mathur, 2017).
3. **BATADAL (BATtle of the Attack Detection ALgorithms) dataset:** simulated C-Town water distribution network, generated via EPANET; no network-layer data. Publicly released through the BATADAL detection-algorithm competition (Taormina et al., 2018).

All datasets are publicly available for research use. No personal data is used, so data protection risks are not expected.

Please describe what skills you have learnt during your course that you will need to use during your project. Also describe which skills you will need to develop in order to successfully complete your project.

Skills developed during the MSc programme relevant to this project:

a) Machine Learning: Random Forest, Isolation Forest, neural networks (from AI and Intelligent Systems module and coursework including the CIC-IDS2017 AI-IDS pipeline).
b) SIEM and network security monitoring: Elastic Stack 8.11, KQL, Fleet agents, MITRE ATT&CK mapping.
c) Python programming: data manipulation, pandas, scikit-learn, matplotlib, SHAP
d) Cyber risk assessment: NIST SP 800-30, MITRE ATT&CK framework
e) Privacy and compliance: GDPR, DPIA, UK data protection law

Skills to be further developed during the project:

a) Deep learning for time-series anomaly detection: LSTM autoencoders, CNN classifiers for network flow data.
b) OT/ICS-specific protocol analysis: Modbus, DNP3, OPC UA feature extraction.
c) XAI implementation: SHAP TreeExplainer and FastSHAP for network traffic classification models.
d) - AHP-Entropy weighting methodology for composite metric design and RAS derivation

Time and Resource Planning

Will Standard Departmental Hardware be used? YES/NO

NO

If NO please outline the Hardware/Materials to be used:

Personal laptop (ASUS Vivobook, Intel Core i7, 16GB RAM, Ubuntu 22.04/Windows 11 dual boot) for running Python ML pipeline, Jupyter notebooks, and virtual environments. Google Colab Pro for GPU-accelerated LSTM/CNN training where local resources are insufficient.

Will Software which is already available in department be used? YES/NO

NO

If NO please outline the Software to be used including how any necessary licences will be obtained:

Primary software: Python 3.11, VS Code, Jupyter Notebook, scikit-learn, TensorFlow/Keras, SHAP, pandas, numpy, matplotlib, seaborn, and scipy. These will be used for preprocessing, feature engineering, model training, evaluation, visualisation, and RAS computation. Google Colab Pro will be used as a fallback for GPU-accelerated training if needed.

Google Colab Pro may be used for GPU-accelerated training if local resources are

insufficient.

Will the project require any Programming? YES/NO

YES

If YES please list the (potential) Programming Languages to be used (including any IDEs and Libraries you may make use of):

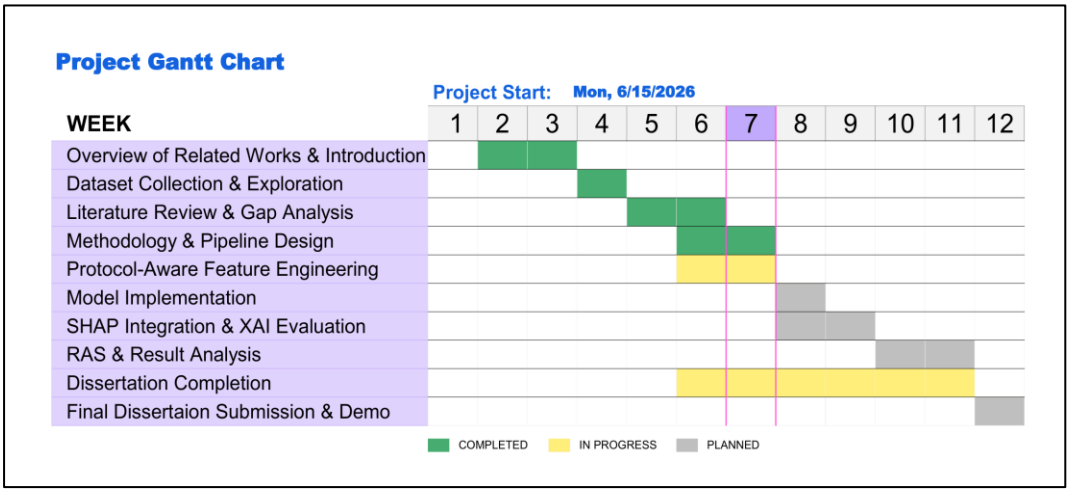
Python 3.11: This will be the primary language for data preprocessing, feature engineering, model training, evaluation, and XAI. **IDE:** VS Code with Python extension; Jupyter Notebook for exploratory analysis. **Key libraries:** scikit-learn (Random Forest, Isolation Forest, feature selection), TensorFlow/Keras (LSTM autoencoder, CNN), SHAP (TreeExplainer, FastSHAP), pandas/NumPy (data manipulation), matplotlib/seaborn (visualisation), SciPy (AHP-Entropy weighting).

Table of Risks (*if non Standard Hardware and/or Software to be used please also include backup options/ contingency plans here)*:

Risk-id Description	Probability/Likelihood of occurring	Best practice prevention measures	Remedy
R1: Dataset unavailability or format change (SWaT, WaDi or BATADAL may become temporarily inaccessible)	Low	Download and store datasets locally at project start. Verify checksums.	Switch to alternative confirmed OT dataset (e.g. BATADAL or EPIC). Pipeline is dataset-agnostic by design.
R2: Insufficient computational resources for deep learning (LSTM/CNN training).	Medium	Use Google Colab Pro for GPU-accelerated training. Profile memory usage early.	Reduce model complexity or batch size. Use pre-trained feature extractors. Prioritise Random Forest/Isolation Forest as fallbacks.
R3: Poor model performance on OT datasets due	Medium	Apply SMOTE oversampling and weighted	Adjust class weights, use anomaly-score

to class imbalance.		loss functions. Use F1 as primary metric not accuracy.	thresholding, or switch to one-class classification approach.
R4: Project scope creep delaying RAS implementation.	Medium	Prioritise core pipeline and RAS over optional extensions.	Descope CTI integration to future work. Maintain core five-phase deliverables as non-negotiable.

Gantt Chart (must include milestones and deliverables):



Project Plan References

Please include a list of References used in this Plan (using Harvard reference style):

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Submission Date:

6th July 2025

PLEASE NOTE THAT SHOULD YOUR PROJECT UNDERGO ANY MAJOR CHANGES FOLLOWING THE SUBMISSION OF THIS PLAN YOU ARE EXPECTED TO SUBMIT AN UPDATED PLAN WHICH ACCURATELY REFLECTS YOUR PROJECT.

MSc Project's GDPR and Ethics Checklist 2026

STUDENT NAME	Toochukwu Praise Ajoku
STUDENT NUMBER	2303 2201 01
E-MAIL	y5b76@students.keele.ac.uk
MODULE CODE	CSC-44120
PROJECT OR WORK PLACEMENT TITLE	AI-Enhanced Network Traffic Analytics for Threat Detection and Cyber Resilience Metric Optimisation (OT/ICS Focus)
KEELE SUPERVISOR NAME	Dr Mohamed Ghanem
DATE	8th July 2025

Complete all the questions below and electronically sign and date at the end. Ask your supervisor to check the responses and to also electronically sign and date below. You should submit this completed checklist to the KLE drop-box provided.

Please retain your own copy of the completed checklist as all end of project reports/dissertations must provide a copy of this completed checklist in the Appendices. If a Project Ethical Review Application Form has also had to be completed a copy of the final ethical approval certificate must also be included with your Project Plan document in your final report/dissertation Appendices.

GDPR Check

Does your project involve the use or collection of "personal data" for which permission will not have been explicitly granted? **(NOTE:** Before answering, **please read and carefully consider the GDPR Check Guidance at the end of this form).**	No

Ethics Check

Will your project involve the use of human participants or human data? (Before answering, **please read and carefully consider the Significant Ethical Concern Checklist, below**). Please Note: software evaluation by any other person counts as human participation. If you ask people their opinions about software or use their data in any way, you need to seek ethical approval first. If you answered "Yes" to this question you must discuss your plans with your supervisor and complete the Computer Science Project Ethical Review Application Form and prepare a Participant Information Sheet and Consent Form using the templates that can be found at the end of that application form.	No

Significant Ethical Concern Checklist

Could the project expose the participants or the project student to images and/or information that they might find distressing (e.g. pictures or descriptions of	No

injuries, symptomatic health conditions, or atrocities, or pictures or descriptions of tumours or cancerous cells, or creatures in distress)?	
Does the project involve deception of the participants?	No
Could the project uncover information about identifiable individuals that could cause embarrassment or distress to one or more of those individuals (e.g. evidence of illegal or unethical behaviour, such as fraud or illegal drug use or a personal revelation)?	No
Could the project cause pain, discomfort or risk to the participants and/or the project student?	No
Will the project involve participants who are vulnerable in any way? (e.g. participants who are under 18, or who are mentally or physically impaired, or participants who may feel under pressure to participate.)	No
Does the project involve recall or discussion of personal or sensitive memories?	No
Does the project involve a significant risk of participants later regretting taking part?	No
Does the project involve procedures which are likely to provoke interpersonal or inter-group conflict?	No

If the answer to any of the Significant Ethical Concern Checklist questions is "Yes" (or you think "Maybe") you must discuss your project and aims with your supervisor and possibly with the Module Co-ordinator to assess whether an appropriate level of ethical scrutiny might be required via the completion of the *Faculty of Natural Sciences (non-Psychology) Research Ethics Application Form*.

GDPR Check Guidance:
Personal data includes any and all of: names, addresses, emails, phone numbers, bank details, employment details, IP addresses, date of birth, medical or health data, images, video or audio recordings.
If you answered "Yes" you must not proceed with your project.
It is illegal under European GDPR legislation to make use of personal data without explicit permission. Discuss your project with your supervisor and revise your plans to ensure you do not risk illegal use of personal data.
Note. Even if personal data is publicly available on the Internet, it must not be used without permission. (Also note that you cannot contact individuals to request permission to use their on-line data without prior ethical approval to do so).

It is strongly recommended that you either:

1. use non-personal data for your project, or
2. use existing, well-established, publicly available databases or data repositories, for example: https://archive.ics.uci.edu/ml/index.php, https://physionet.org/ or https://www.kaggle.com/ etc. (A list of acceptable data repositories is maintained on the CSC-30014 KLE pages.) You might also see, for example, https://blog.scrapinghub.com/web-scraping-gdpr-compliance-guide) for further information.

I confirm that the responses are correct and that the project, as proposed, is GDPR compliant and that ethical approval will be sought if required and that any work requiring ethical approval will not take place unless ethical approval has been granted. If, during the course of the project work, any of the information supplied on this checklist changes substantially a new checklist will need to be completed and then brought to the attention of the School's Ethics Advisory team.

Signed (Student) Toochukwu Praise Ajoku	Date: 8th July 2025

I confirm that I have read the form and that the project, as proposed, is GDPR compliant and that, to the best of my knowledge, the ethical information is correct.

Signed (Supervisor) MCGHANEM	Date: 8th July 2025

Electronically typed signatures and dates *are* acceptable.